

PBot Integrative Paleobotany Portal

The community gateway to fossil plant research and education

How to Guide

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What is PBot?

The Integrative Paleobotany Portal (PBot) is a free online software for fossil plant data management, analysis, and exploration. Core to the PBot mission is providing broad access to paleobotanical research tools and data and to provide an efficient pipeline for data entry that ensures that research products and data are freely available to the general public for the long term. PBot was created because plant fossils present unique challenges that are not accommodated by existing paleontological databases. Plant organs (e.g., leaves, wood, fruits, seeds) are rarely preserved attached to each other, and it can be difficult to associate different organs as part of the same plant and to assign formal taxonomic names to the isolated organs. PBot provides the necessary infrastructure and tools for researchers to record a rich array of fossil plant data and to coordinate that data across institutions, researchers, time periods, and geography.

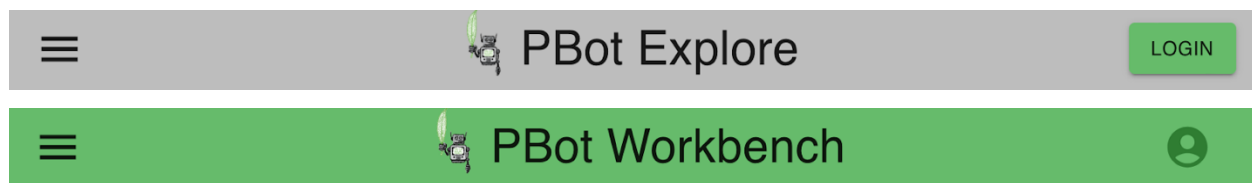
PBot development began in 2021, with funding from NSF EarthCube Award #2026961. The PBot development team committed to creating a system that is “of the community, by the community, and for the community,” and feedback from the international paleobotanical community was incorporated throughout the development process. PBot works with and complements existing databases (iDigBio, the Paleobiology Database, Plant Fossil Names) to enhance utility and accessibility of paleobotanical data, allowing paleobotanists to more easily fulfill NSF (or other funding agency) data management plans.

The PBot Home Page

The Home Page is where you first land when you visit pbot.paleobiodb.org. You can navigate to the different pages and functions either by clicking the green buttons (e.g., “Explore Fossil Plants” or “About”) or by selecting them in the hamburger menu (the three black lines in the upper left corner). Information about PBot and paleobotanical resources are available on the “About”, “How to use PBot”, “Resources”, and “Go to Education & Outreach Hub” pages.



Selecting “Explore Fossil Plants” takes you to **PBot Explore**, our search portal. No login is required, and users can search or browse all publicly accessible data. Some of the most notable data includes fossil specimens and species (or taxa), their descriptions and images, and the location of fossils in time and space. Selecting “Go to Workbench” takes you to the **PBot Workbench**, where registered users can enter paleobotanical data that is either accessible to all or stored in a private group and accessible only to chosen registered users. All private data should be made public upon publication. The PBot page header is gray when in PBot Explore and green when in PBot Workbench. Clicking on the robot icon in the top header will return you to the home page.



How to Register for PBot

While anyone can view public data via PBot Explore, users must register to access the PBot Workbench and to enter, edit, or delete data from the PBot database. To register, click the green “LOGIN” button on the top-right of the page. Then, click “REGISTER” below the login fields.

After registration, PBot Users are automatically added as a Person node based on their registration information. You do not need to enter yourself as a person. If someone else has already created a Person node for you, then any information you add during registration will be added to that Person node.

REQUIRED FIELDS:

Given name – Provide the given name of the individual being entered. Providing the full given name is preferred, but in cases when only an initial is available, do not provide a period after the initial (e.g., M).

Surname – Provide the surname of the individual being entered. When entering suffixes, include a comma before the suffix and period after the suffix (e.g., Surname, Jr.)

Email – Use standard email address formatting including the @ symbol and organization with domain (e.g., name@organization.edu).

Why do you want a PBot account? – PBot uses this field to determine if you are a human. Please enter a quick response for verification (e.g., I am a postdoc and will be entering data with Ellen Currano’s lab group.) Information entered here will not be publicly available

Password – Enter a password you will remember. At this time, we do not support automated password recovery. If you forget your password, please contact us at pbotportal@gmail.com.

Confirm password – Re-enter your password to confirm accuracy.

STRONGLY RECOMMENDED FIELDS:

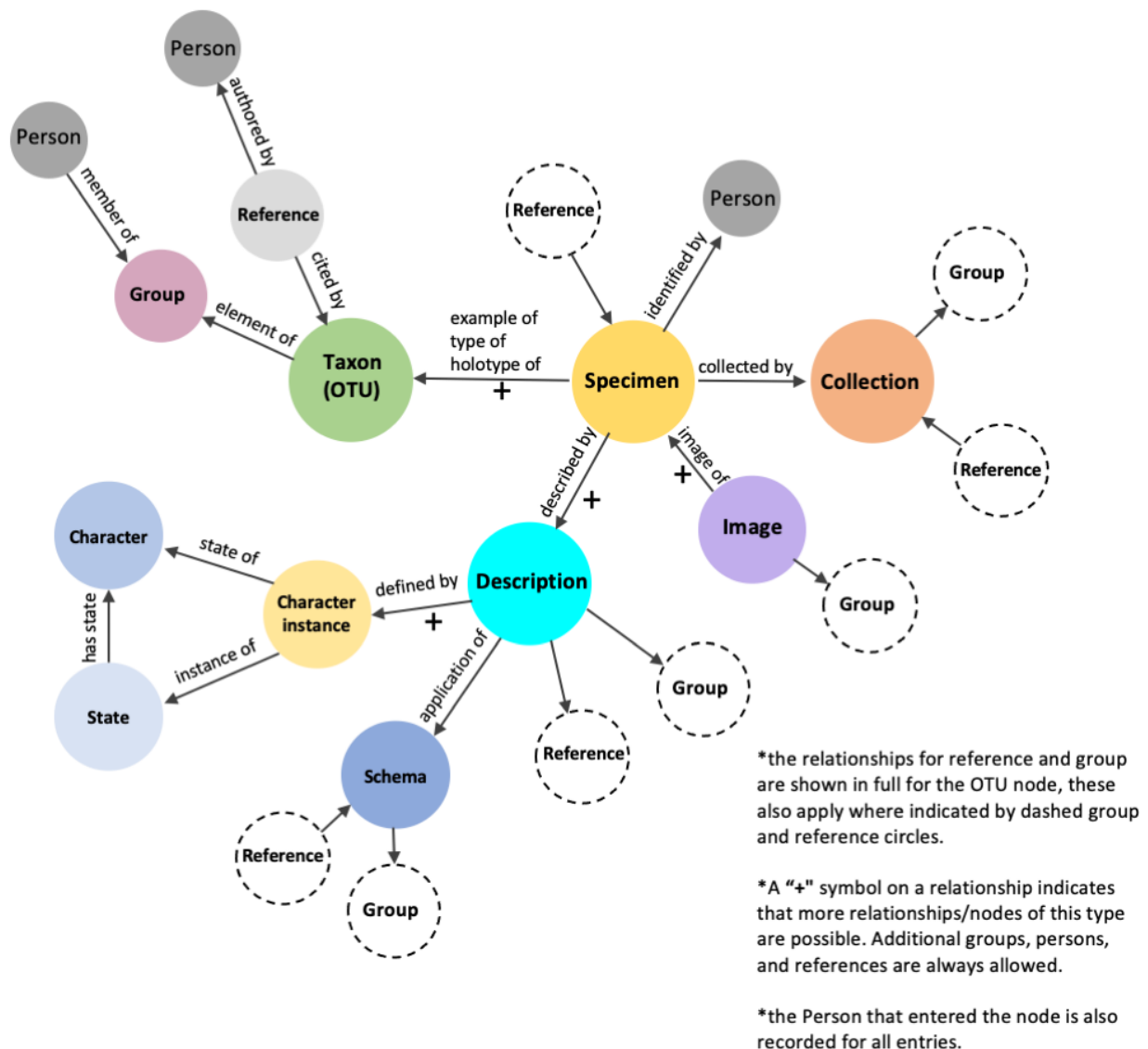
Middle name/initial – Whenever possible, enter the individual’s complete middle name. If entering just an initial, do not include a period (e.g., M).

Short bio – Let the PBot community know who you are! In just a few sentences, tell us about your research goals using keywords so others can find you. This information will be viewable to the public.

ORCID – ORCID IDs are entered as a 16-digit number without hyphens. Formatting will be applied after the user moves to another field or clicks submit.

A Brief Overview of the PBot Architecture

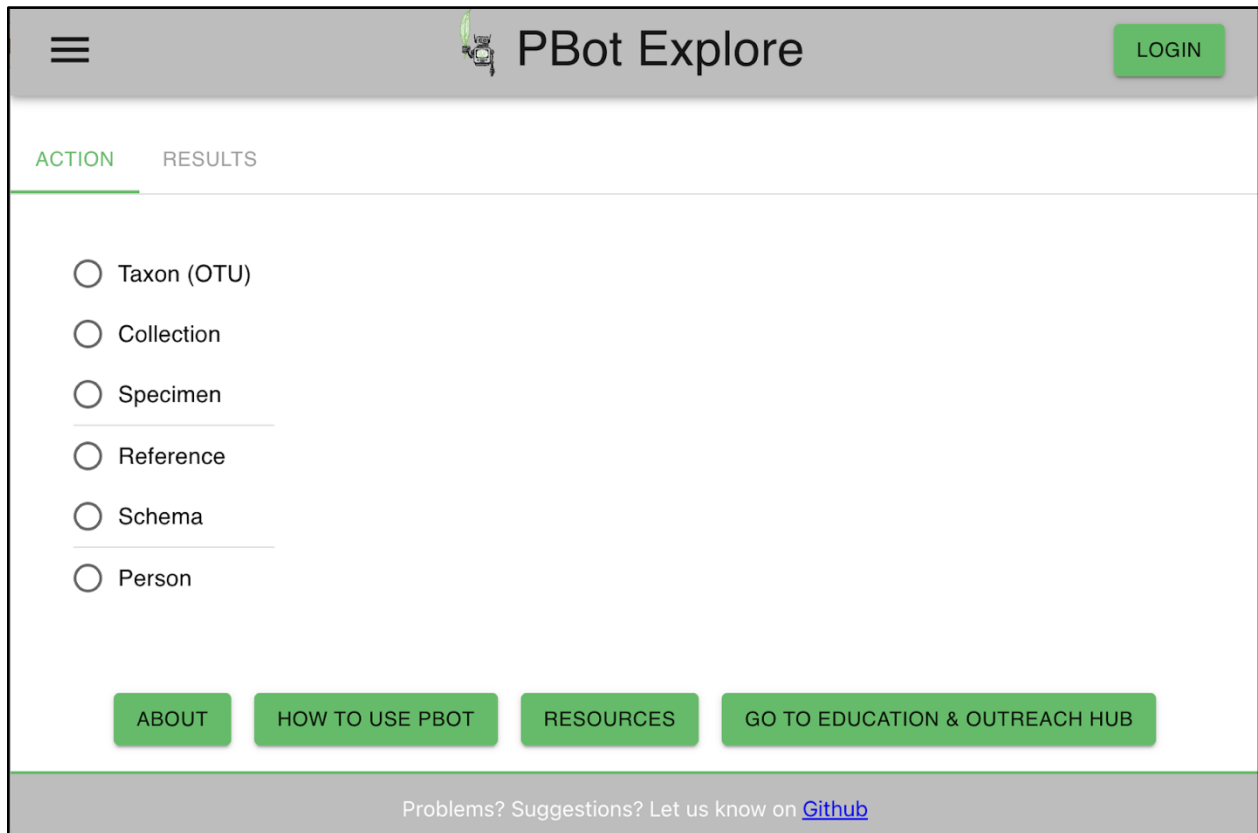
PBot utilizes a graph database that consists of nodes and relationships. Each data entry form corresponds to a data object called a **“node”** in the PBot database. Nodes store the data entered into each field on a form as properties of the node. Defined relationships between nodes enable information to flow between nodes. The conceptual diagram below shows the full PBot architecture, with nodes as circles and relationships as arrows. The table on the following page provides a summary of the major data nodes and the information held as properties of those nodes.



Node	Explanation & Properties
Person	Identifying information about individuals. Sample uses: author of a reference, identifier of a specimen, describer of a specimen.
Reference	Basic information for published references or, prior to publication, informal references. PBot is not a substitute for publishing in the peer-reviewed literature.
Schema	Collections of Characters (questions- e.g., what is the shape of a leaf?) and States (answers- e.g., ovate) used to describe specific kinds of plant fossils. Example: <i>The Manual of Leaf Architecture</i> .
Collection	The geographical, geological, and sampling data for specimens (and by extension OTUs). Information flows between PBot and PBDB.
Specimen	The data about individual specimens, including repository information, iDigBio and GBIF identifiers, and preservation details.
Taxon (OTU)	Accommodates formally named Linnaean-style fossil taxa and informal taxa such as morphotypes. Data entry fields include name, authority, and diagnosis, as well as various taxonomic information. OTU = operational taxonomic unit.
Description	The container that holds the collection of Character States that describe one or more specimens and 1+ references. If specimens are described separately and later determined to be the same OTU, then descriptions can be “merged” (reversibly) in the OTU’s page.
Image	A photograph or other illustration of a specimen. PBot can display images stored in any digital repository as long as each image has its own stable URL.
Group	Allows users to manage access to other nodes while work is in progress. After publication, users should make nodes public.

Using PBot Explore

From the PBot home page, select the “Explore Fossil Plants” button to search PBot data. First, select whether you would like to search for Taxa (OTUs), Collections, Specimens, References, Schemas, or Persons by selecting the bubbles on the left side of the screen. You will then see the various search parameters for each node type. Click the gray submit button to view search results.



The screenshot shows the PBot Explore web interface. At the top, there is a navigation bar with a hamburger menu icon on the left, the PBot logo and the text "PBot Explore" in the center, and a green "LOGIN" button on the right. Below the navigation bar, there are two tabs: "ACTION" (highlighted in green) and "RESULTS". Under the "ACTION" tab, there is a vertical list of search categories, each with a radio button: "Taxon (OTU)", "Collection", "Specimen", "Reference", "Schema", and "Person". Below these categories, there are four green buttons: "ABOUT", "HOW TO USE PBOT", "RESOURCES", and "GO TO EDUCATION & OUTREACH HUB". At the bottom of the interface, there is a gray footer bar with the text "Problems? Suggestions? Let us know on [Github](#)".

PBot searches utilize the “And” Boolean Operator. In other words, if you enter more than one search parameter, PBot will only return documents that contain all the parameters you entered.

Whenever searching PBot (via PBot Explore or by clicking the magnifying glass when entering data in a form), make sure that you are selecting all data access groups that you want to search. The default is to only search public, and you will not get results from your private groups. Search by group is located in the Metadata Accordion at the bottom of each search page.

With one exception, searches are not case sensitive and will return all items that contain the exact character string entered. For example, if you were to search for Collections by Collection name and entered “mu”, your results would include any Collection name with “mu” in it. The only exception is that the OTU taxonomy fields are case sensitive. Data entry and search

queries should follow standard capitalization procedures (i.e., all validly published names are capitalized except the specific epithet, whereas informal names are lower case).

The search results page displays a table with basic information on all nodes that matched your search terms. You can click on any green text in the search results table to view the full information about that record, which opens as a new tab in your browser window. You can download your search results as a JSON or pdf file using the links under the table.

Introduction to Data Entry

From the PBot home page, select the “Go to Workbench” button to enter or edit PBot data. To begin, use the menu on the left side of the screen to select a node type. A data entry form will now be visible. At the top of the data entry form, you can see buttons to create (plus sign), edit (pencil icon), or delete (minus sign) a node.

The screenshot shows the top portion of a data entry form. At the top left, there are three buttons: a plus sign (+), a pencil icon, and a minus sign (-). Above these buttons are three colored arrows pointing to them: a green arrow pointing to the plus sign, a purple arrow pointing to the pencil icon, and a pink arrow pointing to the minus sign. To the right of these buttons is a blue link that says "Click here to expand accordion and see entire menu" with a blue circular arrow icon pointing upwards. Below the buttons are two sections: "Required fields" and "Optional fields", each with a downward arrow icon. At the bottom left are two buttons: a gray "SUBMIT" button and a green "RESET" button. A blue arrow points from the "SUBMIT" button to a blue link below it that says "Click here to submit the form".

Create a new node: Choosing the create button (plus sign) will bring you to a blank data entry form. All fields in the “Required fields” accordion must be completed to successfully submit the form. “Optional fields” are not required but encouraged. Click the gray submit button when you have completed data entry. If any required fields were skipped or not formatted correctly, you will see a red error message below the field(s) causing the submission error.

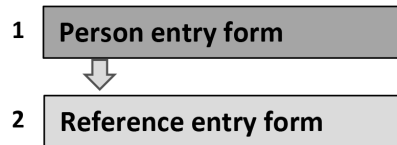
Edit an existing node: Choosing the edit button (pencil icon) will take you to a search field for the node of interest. Once you select a node to edit, you will see the data entry form for that node with all the data that have been entered. Make your edits and then push the gray submit button to save your changes.

Delete an existing node: Choosing the delete button (minus sign) will also take you to a search field for the node of interest. Select the node, wait until the information for that node is displayed, and press the gray submit button to delete the node. Deleting a node is irreversible without technical assistance, so think carefully before you delete. Selecting the Cascade option will delete all other nodes that rely on the node you are deleting. For example, deleting a Collection will also delete all Specimens assigned to that Collection, or deleting a Specimen will also delete all Images associated with that Specimen. If you do not select the Cascade option, you will not be able to delete a node if other nodes rely on it. For example, you cannot delete a Collection if Specimens are assigned to that collection. Person nodes cannot be deleted.

The Data Entry Workflow Overview

The major data elements (the color-coded boxes) are entered in PBot in the order outlined below. You can start at different points in the workflow, or only enter select data elements as needed, as long as any required supporting data (e.g., references) have been entered first. Follow steps A through C. Step D (schema building) is a special case scenario for creating new schemas that enable recording of character states that are not already available in existing schemas. See subsequent pages for specific information for each data element.

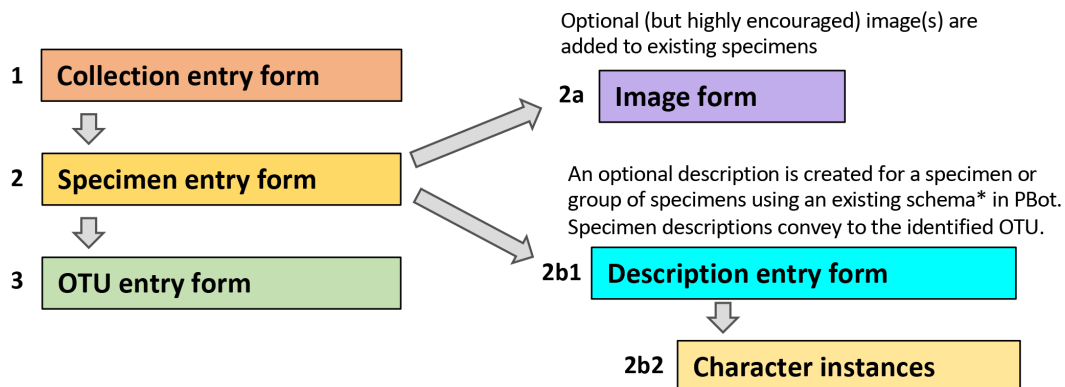
(A) Adding References: Collection, OTU, Schema, and Description nodes all require at least one reference. Start with Person nodes to enter authors and then enter Reference nodes.



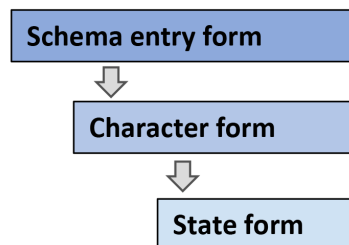
(B) Add data access Group, if needed: Create a data access Group if you would like to enter private data. Only Registered Users can be added to data access Groups, but group members can be edited at any time.

Group entry form

(C) Adding Specimens and Taxa: Dependencies require that a Collection is created before the Specimens that belong to that collection, and that at least one holotype or exemplar Specimen is created before an OTU is created. Follow the entry order 1-3 below; optional steps 2a and 2b1-2 can be entered at any time after a specimen is created.



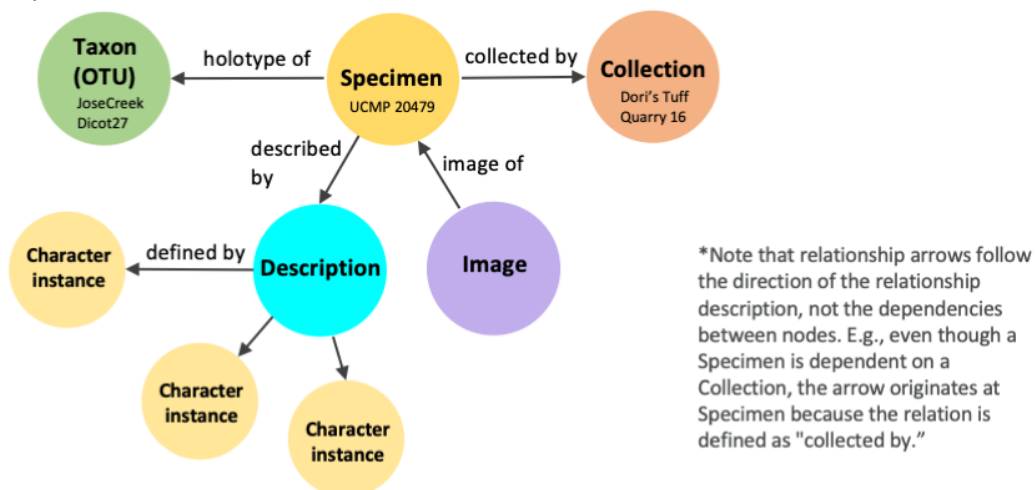
(*D) Adding a new Schema: Make a new schema in PBot for recording observations (as observed states of characters) following the workflow below. The Description form (see C above) allows users to select a schema and record the observed states for specimens.



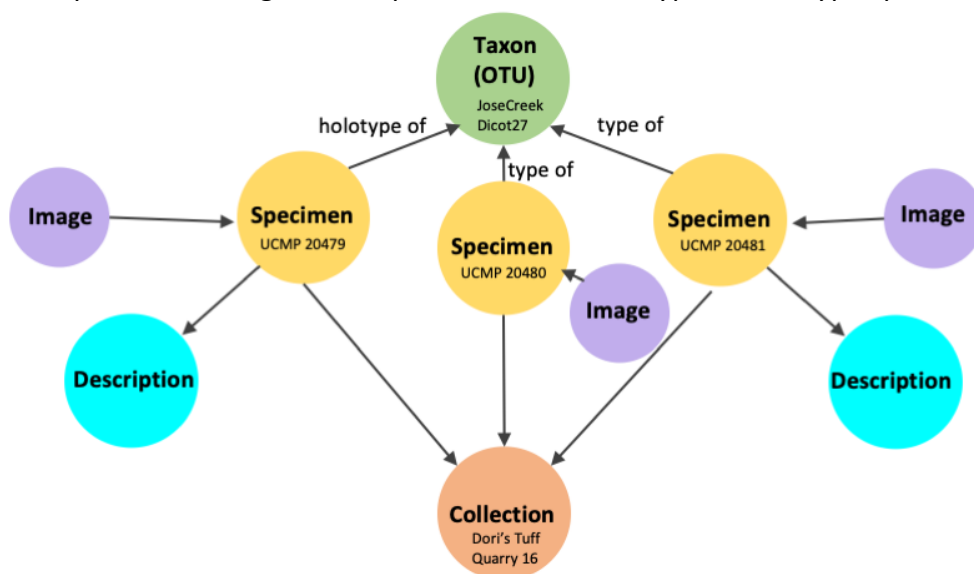
*Entry of characters and their states is flexible as long as the relevant parent character or state is entered first. You can enter all characters before adding states, or enter each character followed by all or some of its states.

Below are a few node-relationship diagrams showing how the major data elements interact for various case scenarios. We start with a simple case scenario and follow with more complex scenarios. We have omitted some supporting nodes (e.g., references) for clarity.

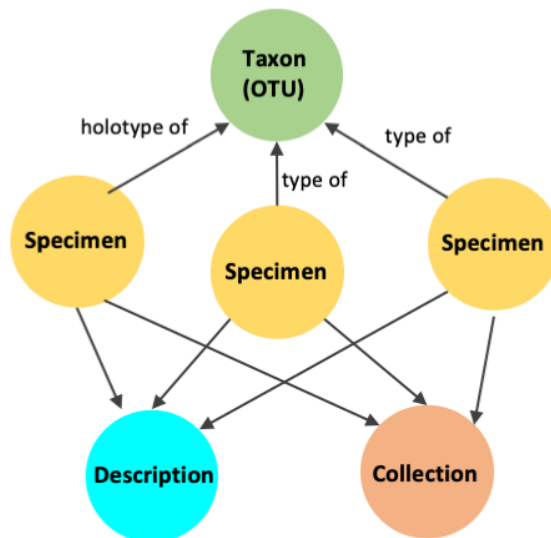
Example 1: An Operational Taxonomic Unit (OTU) was erected based on one described specimen, which is the holotype/exemplar. The OTU inherits the collection data (geologic context, location, etc.), description, and image from the specimen. The description is comprised of all the character states selected from the chosen schema (each character state is a separate character instance).



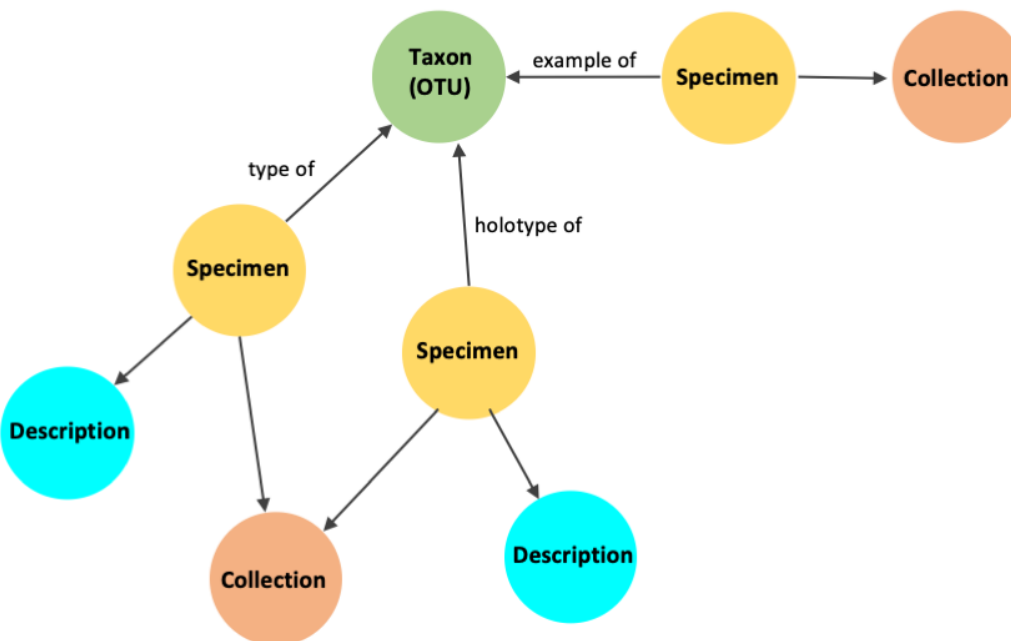
Example 2: An OTU was erected based on several specimens from the same collection. Each specimen has an image, and two of the specimens have individual descriptions (Character instances have been removed for clarity). When viewing the OTU, you can choose to see the holotype description or a merged description from the holotype and all type specimens.



Example 3: Here, the OTU was erected based on several specimens with a shared description (i.e., a description was made based on the combined study of multiple specimens) from the same collection. Note that a merged description and holotype description in this case would be identical. In effect, this is how taxa are described in the literature. Images have been omitted from the diagram for clarity.



Example 4: This OTU was erected based on two specimens from the same collection that were described individually (the type and holotype specimens, as indicated by their relationship to the OTU). Later, a specimen from a different collection was identified as belonging to the existing OTU and added to it as an identified specimen (the “example of” relationship). The collection information associated with the newly added specimen would expand the geographical or temporal range of the OTU.



Person Node

Instructions for data entry

PBot uses Person to record limited, but specific identifying information about individuals. Individuals who will be added to the PBot database include users and published authors. Note, if you are a registered user, a Person Node has already been created for you.

PBot does check internally for duplicates when you submit a new Person Node, but the given name and surname must exactly match the current PBot entry, including capitalization. The best practices to prevent duplicates are to check for existing persons before entering new data and to enter as detailed information as possible to distinguish between individuals with similarities even if that information is not required (e.g., complete middle name).

Recommended process to check whether a person has already been entered in PBot:

- Make a list of all the people you would like to use in your data entry process. For example, create a list of authors for a reference.
- Go to PBot Explore, choose a Person search, and enter the Surname. Determine which members on your list still need to be entered.
- Enter new persons using the Add (+) function in PBot Workbench.

REQUIRED FIELDS:

Given – Provide the given name of the individual being entered. Providing the full given name is preferred, but in cases when only an initial is available, do not provide a period after the initial (e.g., M).

Surname – Provide the surname of the individual being entered. When entering suffixes, include a comma before the suffix and period after the suffix (e.g., Surname, Jr.)

OPTIONAL FIELDS:

Middle name/initial – Whenever possible, enter the individual's complete middle name. If entering just an initial, do not include a period (e.g., M).

Email – Use standard email address formatting including the @ symbol and organization with domain (e.g., name@organization.edu).

ORCID – ORCID IDs are entered as a 16-digit number with hyphens. Formatting will be applied after the user moves to another field or clicks submit.

Reference Node

Instructions for data entry

PBot's goal is for all stored data (Collections, Specimens, OTUs, Descriptions, Schemas, proposed Synonyms) to be associated with a published reference. However, as PBot is also a workbench for paleobotanical research, data can also be linked to informal, unpublished references (e.g., "Ellen's Hanna Basin Research") prior to publication. PBot uses References to ensure data quality, reproducibility, and tracking; however, PBot is not intended to be a substitute for publishing in peer-reviewed literature, nor is it intended to automatically generate comprehensive bibliographies. Note that all Reference nodes are publicly accessible to avoid duplication of records, even if data associated with that Reference is private.

Entering References:

Before you begin entering a Reference, make sure that Person nodes have already been created for each author. Should you realize mid-entry that a Person node does not exist for one of the authors, open PBot in a new browser window and create that Person node. When you return to the partially completed Reference form, this Person node is now available to select. It is also possible to edit the Reference later to add the additional author after creating their Person node.

To enter a Reference, first select the **Publication type** to see the required and optional fields specific to that reference type. Choose from journal article, standalone book (includes theses, monographs, compendiums, as well as other standalone books), edited book of contributed articles, contributed article in edited book, or unpublished.

Prior to creating or editing a Reference (excepting unpublished references), you must determine whether the Reference already exists in the Paleobiology Database (PBDB). To do this, enter data for one or more fields in the PBot Reference form (recommendation: either title, journal and year, or one author), and then click the magnifying glass next to PBDB ID. A new window will appear with PBDB search results. If the Reference already exists in PBDB, you can import the data into PBot by first clicking "Populate all fields", and then clicking on the correct Reference. Note that because author information is stored in PBot as Person nodes, you still need to manually enter authors (see instructions below under the "Author" required field). If the Reference does not appear in the list, click "Cancel" and proceed to enter the remaining fields to complete the new record.

JOURNAL ARTICLE

REQUIRED FIELDS:

Title – Enter the entire article title and do not use abbreviations. If the title is not in English, enter the original title first and then enter an English-language translation in brackets.

Year – The year (include all four digits) in which the article was published.

PBDB ID – If the Reference already exists in the Paleobiology Database, then this field stores the Reference’s PBDB identification number. If the Reference does not exist already in PBDB, this field may be left blank. See instructions at the beginning of this document (above Journal article) for instructions on searching PBDB.

Authors – Link each author’s Person node to the Reference. The “Order” field allows you to enter author order (numbers only). Click the magnifying glass to bring up a PBot search for Person nodes, enter search information, and choose the correct Person. Click “ADD AUTHORS” to search for the next author’s Person node. Make sure to enter all authors.

Journal – Enter the full title of the journal. Do not abbreviate.

Volume – Enter the volume number in which the article was published.

OPTIONAL FIELDS:

Number – Enter the issue number in which the article was published.

First page number – Enter the first page number of the article. If the pages are numbered with combinations of numbers and letters (e.g., “A-32”), put everything in and do not strip the letters. Do not put a hyphen after the page number.

Last page number - Enter the last page number of the article. If the pages are numbered with combinations of numbers and letters (e.g., “A-32”), put everything in and do not strip the letters. Do not put a hyphen before the page number.

Notes – Enter any additional comments you have about the Reference.

DOI – The digital object identifier of the Reference. Only enter the unique ID number - for example, enter “10.1086/663967”, not “https://doi.org/10.1086/663967”.

STANDALONE BOOK

REQUIRED FIELDS:

Title – Enter the entire book title and do not use abbreviations. If the title is not in English, enter the original title first and then enter an English-language translation in brackets.

Year – The year (include all four digits) in which the book was published.

PBDB ID – If the Reference already exists in the Paleobiology Database, then this field stores the Reference’s PBDB identification number. If the Reference does not exist already in PBDB, this field may be left blank. See instructions at the beginning of this document (above Journal article) for instructions on searching PBDB.

Authors – Link each author’s Person node to the Reference. The “Order” field allows you to enter author order (numbers only). Click the magnifying glass to bring up a PBot search for

Person nodes, enter search information, and choose the correct Person. Click “ADD AUTHORS” to search for the next author’s Person node. Make sure to enter all authors.

Publisher – Enter the publisher of the book.

First page number – Enter the first page number of the book (likely, 1). If the pages are numbered with combinations of numbers and letters (e.g., “A-32”), put everything in and do not strip the letters. Do not put a hyphen after the first page number.

Last page number - Enter the last page number of the book, which is also the total number of pages in the book. If the pages are numbered with combinations of numbers and letters (e.g., “A-32”), put everything in and do not strip the letters. Do not put a hyphen before the last page number.

Type – Choose from the following options: thesis (any degree level), monograph, compendium, or other.

OPTIONAL FIELDS:

Notes – Enter any additional comments you have about the Reference.

DOI – The digital object identifier of the Reference. Only enter the unique ID number - for example, enter “10.1086/663967”, not “https://doi.org/10.1086/663967”.

EDITED BOOK OF CONTRIBUTED ARTICLES: *for citing the whole book*

REQUIRED FIELDS:

Title – Enter the entire book title and do not use abbreviations. If the title is not in English, enter the original title first and then enter an English-language translation in brackets.

Year – The year (include all four digits) in which the book was published.

PBDB ID – If the Reference already exists in the Paleobiology Database, then this field stores the Reference’s PBDB identification number. If the Reference does not exist already in PBDB, this field may be left blank. See instructions at the beginning of this document (above Journal article) for instructions on searching PBDB.

Editors – Unlike authors, editors are not associated with Person nodes and should be typed in manually. Do not spell out editors’ full given and/or middle names. Put a period after each initial and a space between each pair of initials (For example: C. C. Cleveland). Use comma separators, and do not use the word “and” before the last editor.

Publisher – Enter the publisher of the book.

First page number – Enter the first page number of the book (likely, 1). If the pages are numbered with combinations of numbers and letters (e.g., “A-32”), put everything in and do not strip the letters. Do not put a hyphen after the first page number.

Last page number - Enter the last page number of the book, which is also the total number of pages in the book. If the pages are numbered with combinations of numbers and letters (e.g., “A-32”), put everything in and do not strip the letters. Do not put a hyphen before the last page number.

OPTIONAL FIELDS:

Notes – Enter any additional comments you have about the Reference.

DOI – The digital object identifier of the Reference. Only enter the unique ID number - for example, enter “10.1086/663967”, not “https://doi.org/10.1086/663967”.

CONTRIBUTED ARTICLES IN EDITED BOOK: *for citing an individual article*

REQUIRED FIELDS:

Chapter title – Enter the entire title of the chapter/article and do not use abbreviations. If the title is not in English, enter the original title first and then enter an English-language translation in brackets.

Year – The year (include all four digits) in which the book was published.

PBDB ID – If the Reference already exists in the Paleobiology Database, then this field stores the Reference’s PBDB identification number. If the Reference does not exist already in PBDB, this field may be left blank. See instructions at the beginning of this document (above Journal article) for instructions on searching PBDB.

Authors – Link each author’s Person node to the Reference. The “Order” field allows you to enter author order (numbers only). Click the magnifying glass to bring up a PBot search for Person nodes, enter search information, and choose the correct Person. Click “ADD AUTHORS” to search for the next author’s Person node. Make sure to enter all authors.

Book title – Enter the entire title of the book and do not use abbreviations. If the title is not in English, enter the original title first and then enter an English-language translation in brackets.

Editors – Unlike authors, editors are not associated with Person nodes and should be typed in manually. Do not spell out editors’ full given and/or middle names. Put a period after each initial and a space between each pair of initials (For example: C. C. Cleveland). Use comma separators, and do not use the word “and” before the last editor.

Publisher – Enter the publisher of the book.

First page number – Enter the first page number of the article/chapter. If the pages are numbered with combinations of numbers and letters (e.g., “A-32”), put everything in and do not strip the letters. Do not put a hyphen after the first page number.

Last page number - Enter the last page number of the article/chapter. If the pages are numbered with combinations of numbers and letters (e.g., “A-32”), put everything in and do not strip the letters. Do not put a hyphen before the last page number.

OPTIONAL FIELDS:

Notes – Enter any additional comments you have about the Reference.

DOI – The digital object identifier of the Reference. Only enter the unique ID number - for example, enter “10.1086/663967”, not “https://doi.org/10.1086/663967”.

UNPUBLISHED

Ideally, an unpublished reference should be a placeholder for an intended future publication. Reference information can be edited at any time; therefore, once your manuscript is published, you can simply edit the Reference fields to match the final publication type and information (in other words, you will not need to individually edit every record in your publication dataset to change to the published version of the reference). For general entry of private data that does not yet have a designated intended publication, you can generate a Reference titled “[Username]’s Personal Workbench”, which serves as a generalized interim private workspace, or one titled appropriately for a private working Group in which you are collaborating with other users. In some cases, you may be entering data that you intend to be public even though it is not in the published literature (for example, museum specimens that are already available online and not currently being studied); for these cases, you can enter a reference to the institution and/or collections database from which the specimen information was derived.

REQUIRED FIELDS:

Title – Keeping in mind that all References are publicly viewable, enter an informative title that matches your comfort-level. Example titles include “SiteName flora project”, “Username’s SiteName research”, “Username’s TaxonName research”, “Username’s Personal Workbench”.

Year – Enter the current year (include all four digits) as a timestamp of when the unpublished reference was created.

Authors – Link each author’s Person node to the Reference. The “Order” field allows you to enter author order (numbers only). Click the magnifying glass to bring up a PBot search for Person nodes, enter search information, and choose the correct person. Click “ADD AUTHORS” to search for the next author’s Person node. Make sure to enter all authors.

Description – A short summary of the unpublished reference. Remember that all Reference nodes are publicly accessible.

OPTIONAL FIELDS:

Notes – Enter any additional comments you have about your unpublished reference, keeping in mind that this will be publicly accessible.

Group Node (Data Access)

Instructions for data entry

PBot uses groups to restrict access to private, unpublished data and to facilitate collaboration between colleagues on private datasets. Only group members will be able to view, add, edit, or delete data entered as private data by the group. Group members can create private data when adding collections, descriptions, taxa (OTU), images, and schemas. Specimens inherit the group status (private or public) from the collection to which they belong.

Because only registered users can be added to a group, be sure all the individuals you would like to add to your group are registered as PBot users. However, members can be added and edited at any time, so you can create a group with only yourself as a member to get started, or for solo work.

REQUIRED FIELDS:

Name – A unique identifier for the group. Group names should be short and specific to the purpose of the group (e.g., Bighorn Basin P-E).

Purpose – Provide a brief description of the group's purpose so that users are clear on what data should and should not be included in this private data repository.

Members – You are automatically included in any groups you create. If you would like to add other registered users to the group, click “**ADD MEMBERS**”, and then use either the dropdown menu or magnifying glass to add people. Continue to add members until your group is complete. If you do not see the user you are searching for in the selection options, it means they have not registered for PBot (even if they have been entered as a Person in PBot).

OPTIONAL FIELDS—not applicable

Collection Node

Instructions for data entry

PBot uses the concept of Collections, as implemented and required by the Paleobiology Database (PBDB), in order for our databases to exchange information. Collections are how PBot records the geographical, geological, and sampling data for specimens (and by extension, taxa). **We can think of a “Collection” as a sample of fossils from a distinct location and stratum that was collected in a known time-frame using a particular method.** Each specimen in PBot must be linked to *one* Collection. See the general comments below for important information on delineating collections.

General comments on Collections: *This section is very important. Please read all of it very carefully. We follow the policies and guidelines established by the PBDB, who provided much of the text below. For more information about collections in PBDB, see <https://paleobiodb.org/public/tips/tips.html>.*

- A geographic locality or stratigraphic level may include many Collections. For example, if fossils were collected from multiple quarry holes at a locality, whether that be from multiple stratigraphic horizons or from laterally sampling one fossiliferous bed, each quarry hole should be a unique Collection. Likewise, if two papers describe material clearly, entirely collected at completely different times from the same place, make two Collection records. We can always lump such records later when we do analyses; we cannot separate out such collections if they have been lumped during data entry. *Lumping collections destroys data.*
- However, *don't* split every paper as a new Collection record. A "Collection" is any set of fossils whose exact geographic and stratigraphic position and date of collection cannot be distinguished, regardless of when they were described, and by whom. So, if five people describe five new, unrelated species/morphotypes in five different papers, don't make five Collection records unless different physical collections from different places and/or strata are involved.
- *Never* make a new Collection record just because you (or your research team) did not create the existing record for the same actual hole in the ground. Instead, add specimens to the existing Collection record.
- The Collection record represents our best current knowledge about the Collection. So, if the original paper said "Eocene" but a later paper narrowed the age to "Middle Eocene," enter "Middle Eocene" as the time interval. Both references need to be associated with the Collection, which you do by adding the new Reference to the Collection.

Prior to creating a new Collection node, please determine whether the Collection already exists in the Paleobiology Database. From the PBot Collections Form, enter the latitude and longitude of the Collection, and then click the magnifying glass next to the PBDB ID field. If the Collection exists in PBDB, first select “Populate all fields,” and then click on the Collection to autopopulate the data into the PBot Collections Form. Enter any additional collection information in the PBot Collection Form and then submit.

REQUIRED FIELDS:

Name – A unique identification that may also aid in recognizing close links among Collections. 80 character limit. For example, if you are entering a series of consecutive beds, you could provide a location name and meter level (e.g., “Saddle Mountain 1.5 meter bed” and “Saddle Mountain 2.0 meter bed”). Keep it short and informative, but do not include the stratigraphic age, location details, or the name of the formation, country, state, etc. We recommend using the location name along with locality number/name, plus any other specifier needed to differentiate from other similar collections. If including a formal locality number, please also include the institution code that generated the locality number to avoid duplicates.

Collection type – Characterize the intent or subject area of the *part of the original publication* from which the Collection was compiled. Be careful because different parts of one paper might have different purposes. When in doubt, use the category that best describes the whole paper.

- **Archaeologic** means collected for archaeologically focused research.
- **Biostratigraphic** means described for the purpose of temporal correlation, or in the course of a geological study (e.g., lists given within descriptions of measured sections).
- **Paleoecologic** means described for the purpose of paleoecological analysis, including taxonomic lists that are given without descriptions of fossils and without discussions of temporal correlation.
- **Taphonomic** means described for the purpose of a taphonomic analysis.
- **Taxonomic** means a systematic paleontology treatment, including simple descriptions of fossils. Use this if all the taxa are included in a systematic paleontology section, or if the taxonomic list is given only in support of an otherwise taxonomic account of one or a few taxa.
- **General faunal/floral** means there is no detailed discussion of the age, environment, ecology, preservation, or morphology of the fossils, but a taxonomic list is given. Use this category only if the other ones are too specific.

Latitude and Longitude – The present-day latitude and longitude (as opposed to paleolatitude or paleolongitude) in decimal degrees. All coordinates must use the WGS84 coordinate system. Latitude values range from -90 to 90, inclusive, with positive values north of the Equator and negative values south of it. Longitude values range from -180 to 180, inclusive, with positive values east of the Greenwich Meridian and negative values west of it.

Please enter coordinates to the highest degree of precision possible, in accordance with rules, regulations, and courtesies regarding fossil locality data. It is your responsibility to determine the sensitivity of your data and enter coordinates accordingly. The Society of Vertebrate Paleontology recommends reporting to 0.1 degrees for high sensitivity, 0.01 degrees for medium sensitivity, 0.001 degrees for low sensitivity, and as many decimal points as possible for non-sensitive data (see full SVP recommendations [here](#)). In general, a precision of no greater than 0.1 decimal degrees latitude/longitude for highly sensitive geospatial data is recommended as a best practice by many scientific organizations, including the U.S. Bureau of Land Management, the Global Biodiversity Information Facility, and museums.

GPS coordinate uncertainty – The radius of uncertainty, in meters, for the GPS coordinates entered. Coordinates may be uncertain due to accuracy-level of a handheld GPS (5 m uncertainty), inability to precisely locate a historic collection (e.g., only a county or township, range, and section are given), or the need to report lower resolution coordinates by law or request of a property owner. For reference, if you are reporting coordinates to 0.1 degree resolution, your radius of uncertainty would be 5550 m.

We have included additional resources for latitude, longitude, and coordinate uncertainty at the end of this cheat sheet.

Country – Select the present-day country in which the Collection occurs from the dropdown menu.

Protected site – Check this box to indicate that the site location and access to it is restricted. Remember, it is your responsibility to enter GPS coordinates at the appropriate resolution.

Timescale, maximum, and minimum interval – The zone, age/stage, epoch, or period from which your Collection is drawn. First, select the most precise timescale you can from the “Timescale” dropdown menu (e.g., international epochs rather than international periods). Your choice will populate a drop-down menu for the “Maximum interval” field. After you make a selection, that value will automatically populate the “Minimum interval” field. If a Collection is known to span, or could span, multiple intervals, change the “Minimum interval” entry to the correct minimum.

Lithology – The lithology that makes up most of the exposure (for an outcrop) or part of the core that was studied (for a sample from a core). If it is not clear which lithology dominates, choose the first one listed in the publication. See the Paleobiology Database [lithology tip sheet](#) for explanations of lithologic terms.

Preservation modes – The kind(s) of fossils that are present. Select as many as apply to any of the fossils in the Collection. We define the fossil types as follows:

- *Adpression/Compression*: 2D fossil in which original organic material remains, although often in a geochemically altered state. This category includes carbonization fossils.
- *Body*: remains of the body itself
- *Cast*: none of the original biomineralized or organic tissue preserved; 3D fossil that is a positive image of the original
- *Charcoalification*: near-perfect, often cellular-level, preservation of plant organs (most commonly wood, leaves, or reproductive organs) as charcoal.
- *Coalified*: vegetative material has been converted to coal.
- *Concretion*: Mazon Creek nodules are examples of concretions.
- *Mold/impression*: none of the original biomineralized or organic tissue preserved; Impressions are 2D fossils, and molds are 3D fossils that are the negative image of the original.

- *Permineralization*: 3D infilling of intracellular spaces with carbonates, iron compounds, pyrite, silicates, etc. Coal balls and petrified wood are examples of permineralized fossils.
- *Recrystallized*: 3D fossil in which the original tissue is replaced by a mineral.
- *Trace*: evidence of behavior, such as feeding damage, coprolites, or trackways

Size classes – PBDB requires that collections include which size classes of fossils have been recovered. Your options are > 10 mm, 1-10 mm, or < 1 mm, and multiple options can be selected. If the size range of specimens is not clearly indicated in a publication, use a liberal estimate to select all possible size categories.

Paleobiology Database ID – From the PBot Collections Form, search the Paleobiology Database to determine if your Collection has been entered there. You must do this search in order to submit the Collections Form. The Paleobiology Database ID is PBDB's Collection ID number, and it will be automatically filled in if you click on a PBDB collection in the PBot search results. If the Collection does not exist already in PBDB, this field may be left blank.

References – Select one or more References that describe the Collection. References must already exist in PBot and can include unpublished references (temporary placeholders for personal workbenches and unpublished collections). Multiple References can be associated with a Collection by clicking "**ADD REFERENCE**". Enter the order number for the Reference to indicate its priority in the reference list, with the most relevant publication as order 1.

Public or Group – The default is for a Collection to be Public, as published collections should always be part of the "Public" access group. If you would like to restrict access to an unpublished site, uncheck "Public" and select which one or more of your private groups can access this Collection. Once a record has been made public, it cannot be returned to a private group. Note that protecting the geographic location of a published locality should be done by making your gps coordinates "fuzzy" and checking the "Protected site" box, not by making the whole collection record private.

OPTIONAL FIELDS: GEOGRAPHIC

State/province – A required field if the country is the USA. Elsewhere, its use is also encouraged (e. g., England is a "state" of the United Kingdom, Ontario is a province of Canada).

Scale of geographic resolution – From the drop-down menu, select the resolution that best characterizes your Collection. Err on the side of conservatism: if you know the Collection is from a particular roadcut but don't know if it does or doesn't come from a small excavation within the roadcut, enter "outcrop" instead of "small collection." If no information is given, do not make a selection. Terms are defined as follows:

- *small collection*: from an area < 10 x 10 m, e.g., a single fossil, an individual bulk sample, or a small quarry
- *outcrop*: from an area < 1 x 1 km, e.g., a large quarry, road cut, or wash

- *local area*: from an area < 100 x 100 km, e.g., approximately a one-by-one degree lat/long rectangle or the size of most individual counties in the USA
- *basin*: from any area > 100 x 100 km

Notes on geographic information – Provide any additional information that you think is important with respect to the geographic location of the Collection.

OPTIONAL FIELDS: AGE

Direct date, Numeric maximum age, Numeric minimum age – Use these fields for numerical age estimates based on actual measurements of rocks and fossils in the actual horizons only (i.e., Collection is associated with a specific geochron measurement). *Do not* enter guesstimates or values derived from time scales (e.g., a range derived from the Gradstein et al. boundaries for a global stage). These fields should not be filled in if a collection can only be assigned to a Period, Epoch, Age, or Series. *Do not* enter values that the author does not believe to be reliable (say, because the sample might be contaminated). There are three kinds of dates:

- *Direct date* means a date for the exact horizon yielding the collection. If one single direct date is given you should enter it instead of any maximum and/or minimum dates. Also, if there is a minimum but no maximum or vice versa, it would be better to enter a direct date.
- *Numeric maximum age* could mean one of two things: (1) the oldest of several direct dates (enter the youngest in the minimum field), or (2) the oldest date for the highest underlying stratigraphic horizon. For example, if the fossils are from unit 7 and there are dates for units 4 and 5, enter the youngest date for unit 5.
- *Numeric minimum age* means the youngest direct date or youngest date for the lowest overlying horizon.

Notes on age information – Provide any additional information that you think is important with respect to the age of the Collection.

OPTIONAL FIELDS: GEOLOGIC

Additional description of lithology – Quote the entire description of the lithology if it includes *any* information you can't record in the other fields of this section or in the required "Lithology" field. Even if there is just a word difference like "carbonate" vs. "lime," the description should be quoted in full.

Stratigraphy: Geologic group, formation, member, bed, notes on stratigraphy – Names of stratigraphic units yielding fossils. Always enter the formation name if it is available. Important: Only provide the name of the stratigraphic unit- do not put the terms "group", "formation", "fm.", "member", or "mbr." In these fields. Provide any additional information that you think is important with respect to the stratigraphic delineation of the collection in the "Notes on stratigraphy" field.

Environment – The depositional environment (or paleoenvironment). Select the most exact environmental category you can figure out. Each section of the pulldown includes some

generalized “indet.” terms that you should use only if a more precise determination is not possible. For more information and full definitions of each environmental setting, see the [Paleobiology Database Environmental Tips](#).

Notes on environment – Provide any additional information, including extensive quotes from the paper, that you used to figure out these values. Don't leave this field blank unless the paper very specifically assigned the collection to an environmental category that exactly corresponds to the ones we use.

OPTIONAL FIELDS: COLLECTING

Collectors – The names of the people who collected the fossils, including the Reference's authors if appropriate. Separate names by commas and do not write out "and." Example: *J. Doe, J. Smith, J. Brown*

Collection methods – The methods used to make the collection. Multi-select is allowed, and you should check both "bulk" and "sieve" for wet or dry screen-washed collections.

- *bulk*: the fossils in a set volume of rock, or on an all surfaces of slabs of rock, were identified. This is the appropriate selection for comprehensive census surveys of fossils.
- *sieve*: the fossils were put through a screen or sieve either in the field or in the lab.
- *core*: the fossils came from a core, regardless of whether all material was identified.
- *quarrying*: a trench, pit, or quarry was dug to collect the fossils, but not everything was necessarily identified (if it was, then call it a *bulk* collection).
- *surface (float)*: loose fossils were picked up at the surface.
- *surface (in situ)*: fossils were found embedded in the rock but no extensive quarrying or coring was done to collect them.
- *salvage*: some fossils, but not all, were collected during an excavation intended for some other purpose (e.g., construction).
- *anthill*: fossils were picked from the debris on the tops of anthills.

Notes on collection methods – Provide as much additional information as possible. Examples: only dicot leaves were identified; specimens were collected from a spoil pile; the quarry was discovered by Doe and excavated by Smith and Brown; the USNM museum collection was made in 1940 and the AMNH museum collection in 1941.

EXTENDED INSTRUCTIONS FOR ENTERING LATITUDE & LONGITUDE:

Decimal degree digits – What digits mean in latitude and longitude decimal degrees using WGS84. The decimal degree explanations below were adapted from [StackExchange](#) on 08-15-2023. Please note the phrase “up to” in these explanations refers to variations in decimal degree distances around the globe. The explanations here refer to the maximum distance represented by a decimal degree digit. To correct for variation and obtain high precision in decimal degree distance, see latitude and longitude at the end of this document.

- The **tens digit** gives a position up to about 1,000 kilometers. It gives us useful information about what continent or ocean we are on.

- The **units digit** (one decimal degree) gives a position up to 111 kilometers (60 nautical miles, about 69 miles). It can tell us roughly what large state or country we are in.
- The **first decimal place** gives a position up to 11.1 km: it can distinguish the position of one large city from a neighboring large city.
- The **second decimal place** gives a position up to 1.1 km: it can separate one town from the next.
- The **third decimal place** gives a position up to 110 m: it can identify a large agricultural field or institutional campus.
- The **fourth decimal place** gives a position up to 11 m: it can identify a parcel of land. It is comparable to the typical accuracy of an uncorrected GPS unit with no interference.
- The **fifth decimal place** gives a position up to 1.1 m: it distinguishes trees from each other. Accuracy to this level with commercial GPS units can only be achieved with differential correction.
- The **sixth decimal place** gives a position up to 0.11 m: you can use this for laying out structures in detail, for designing landscapes, building roads. It should be more than good enough for tracking movements of glaciers and rivers. This can be achieved by taking painstaking measures with GPS, such as differentially corrected GPS.

Degrees, minutes, and seconds (DMS) digits – What digits mean in DMS versus decimal degrees using WGS84. The DMS explanations below were adapted from [USGS.gov](https://www.usgs.gov) on 10-10-2023. The guide provided can be used to conserve the precision of latitude and longitude reported using DMS to the same level of precision using decimal degrees. Consider all contextual information when estimating radius of uncertainty.

- One degree of **latitude** equals approximately 110,947 m, one minute equals approximately 1850 m, and one second equals approximately 31 m.
- One degree of **longitude** equals approximately 87,843 m, one minute equals approximately 1463 m, and one second equals approximately 24 m.
- To convert latitude and longitude precision from DMS to decimal degrees, consider the following guide. Use the DMS column on the left to determine how many decimals to report using decimal degrees.

DMS	decimal degrees
0° 0'	0.0
0° 0' 0"	0.01
0° 0' 0.0"	0.001
0° 0' 0.00"	0.0001
0° 0' 0.000"	0.00001
0° 0' 0.0000"	0.000001

Latitude – Latitude distances are fairly consistent around the globe. The distance of one degree of latitude varies from 111.964 km at the poles to 110.574 km at the equator, a difference of 1,120 m. A [calculator](#) to constrain the exact distance of one degree of latitude or longitude at any latitude is provided by the U.S. Government's National Geospatial-Intelligence Agency (NGA).

Longitude – Longitude distances vary more substantially. One degree of longitude varies from 0.00 km at the poles to 111.320 km at the equator, a difference of 111,320 m. A [calculator](#) to constrain the exact distance of one degree of latitude or longitude at any latitude is provided by the U.S. Government's National Geospatial-Intelligence Agency (NGA).

Specimen Node

Instructions for data entry

Specimen nodes track data about individual specimens and can be linked to the Operational Taxonomic Unit to which they are identified. The geographical, geological, and sampling information about specimens are recorded in the Collection to which a Specimen belongs. Please be sure the relevant Collection has been entered or imported in PBot before entering a Specimen (see Collection instructions). Specimens will “inherit” the privacy status (i.e., whether they are public or in a private access group) from the status of the collection to which they belong. Any Descriptions or Images attributed to a Specimen can be private, regardless of whether the Specimen is public.

PBot does check internally for duplicates when you submit a new Specimen Node, but the Specimen Name must exactly match the current PBot entry, including capitalization. The best practices to prevent duplicates are to follow best practices for Specimen Names (see below) and to check existing Specimens before entering new data.

REQUIRED FIELDS:

Specimen number – Provide a unique name for the Specimen. In most cases, this should be the specimen’s catalog number at its permanent repository. Include the repository code at the beginning of the catalog number if it is not already included in the specimen number (e.g., USNM PAL777113). Do not forget to include the repository code, as different institutions may use the same numbering system and Specimen numbers must be unique in PBot. Institutional/repository codes registered with the Global Registry of Scientific Collections (GRSciColl) can be found at www.gbif.org/grscicoll/institution/search. If a catalog number has not been assigned, a field number or other tracking may be used until the specimen is cataloged. Names can be edited at a later time; it is recommended to record any prior or temporary numbers in the Notes field.

Collection – Choose the Collection to which the specimen belongs. If the Collection for the specimen does not exist yet in PBot, open a new browser and enter a new Collection; after submitting the new collection, use the search tool to find the collection and select it. See Collection instructions for more information.

Parts preserved – Select the organ(s) or part(s) of the plant that are represented by the preserved specimen.

Preservation mode – Select the modes of preservation that are present in the fossil specimen. You may select more than one if needed.

Repository – Enter the full name of the institution at which the Specimen is housed. If the Specimen is under study at a different institution than its intended permanent repository, you

may enter the intended repository if known, or temporarily the institution where the specimen is under study. This should be updated if the Specimen is later transferred.

OPTIONAL FIELDS: OTHER

Notable features preserved – Choose from the list any notable aspects of the specimen that are known to be preserved. This information can help workers know whether a specimen has/preserves certain key features that can be studied, even if not described yet.

References – Choose an existing Reference, or References, for the Specimen if it has been included in any publications. You may choose a personal ‘working reference’ (see Reference Instruction sheet) if the specimen has not been published but is part of an in-progress project. Enter the order number for the Reference to indicate its priority in the reference list, with the most relevant publication as order 1.

Identified by – Choose names of people (Person nodes) that are responsible for identifying the specimen as a particular OTU. Note that Specimens are assigned to OTUs through the OTU page.

Other repository link – Type or paste a url to an existing record for this specimen on any other online database, such as a museum’s online catalog. Multiple entries can be separated by a comma.

Notes – Enter notes about the specimen that the enterer deems necessary to record. This is the appropriate place to discuss the history of the Specimen name (or specimen number), such as providing a field number or previous repository number if a Specimen has been moved. It is particularly important to do this if a publication refers to that Specimen using a different number than the Specimen name. This field can be used for research notes or memos to your collaborators, and you may edit or delete such notes at any time.

OPTIONAL FIELDS: IDIGBIO

This tab provides a lookup function and link to records in iDigBio. Type the Institution code and Catalog number, or the UUID, then click the search icon to find and select the iDigBio record for the specimen.

Institution code – type the institution code of the repository.

Catalog number – type the catalog number of the specimen at the repository. Click anywhere outside of the field and then click the green magnifying glass icon to search for specimen record in iDigBio.

UUID – type the Universal Unique Identifier (UUID). Click anywhere outside of the field and then click the green magnifying glass icon to search for specimen record in iDigBio.

Image Node

Instructions for data entry

PBot uses the Image node to link or store specimen images. **As we do not have unlimited storage space, we strongly encourage you to include stable links to specimen images rather than uploading images directly to PBot.** See the Resources page on the PBot site for some suggested digital repositories.

REQUIRED FIELDS:

Collection – Choose the Collection to which the imaged specimen belongs. You may either search Collections using the magnifying glass or select a Collection from the dropdown menu.

Specimen – Select a Specimen from the dropdown menu to associate an image with a specimen from the selected collection.

Image: File or Link –

To associate a link to an image with the specimen, click the Link button. Paste the URL into the dialogue box and click the Enter button.

To upload an image file to associate with the specimen, click the File button. Select the appropriate file from the dialogue box and click the Upload button. PBot supports .jpg, .png, .svg, .tiff file types. The maximum allowable file size is 15 MB, and we encourage users to keep images as small as possible to conserve server space.

Category – Select an image priority based on the specimen type and what is shown in the image. This selection determines how early the image will be presented in the image carousel on the Specimen and/or OTU page. Images display in the following category order: holotype/exemplar, other type specimen, shows key feature, no priority. Select “holotype/exemplar” or “other type specimen” for images that show the entire specimen (or, in the case of material like cuticle, a representative section), as these will be the first viewed in the carousel. Use “shows key feature” for zoomed-in images of type material or for images of non-type specimens that preserve features that may not be well-exhibited by the type specimens.

Credit – Provide image credit suitable for publication that acknowledges the photographer, repository, and sources as well as any image use limitations such as copyright status.

Caption – Include a brief description of why this image was included with the specimen record, such as image quality and key features within the image.

OPTIONAL FIELDS—not applicable

Description Node

Instructions for data entry

A Description node serves as a container that holds character states that describe one or more Specimens. The ability to describe specimens in aggregate or as individual specimens provides the user with the flexibility needed for different methodologies and for entering published data for which specific specimen-level information is not known. If specimens are described separately (i.e., they each have separate Description nodes completed) and are determined later to belong to the same OTU, then the descriptions can be “merged” (reversibly) in the OTU’s page. The ability to track specimen-level descriptions can also be a useful tool for assessing variability in character states, preservation, or the later removal/ addition of specimens to an existing OTU record.

Workflow for making a description:

1. Generate a Description node. Here you will choose the Specimens being described and the Schema to be used for description. Submit.
2. Select the Edit Description function and select your Description from the list.
3. Now you should see there is a “Character instances” accordion. The accordion is populated with the characters from the schema you selected. Open the accordion to see the character hierarchy of the schema and to submit character states!

REQUIRED FIELDS:

Name – Provide a unique and informative name for the Description. For published OTUs/Specimens, we recommend the format “OTUName/SpecimenNumber + Author(s) + SchemaName + Published Description” (ex: *Davidia antiqua* Manchester MLA Published Description). For unpublished work or new descriptions of published OTUs/Specimens, we recommend the format “OTUName/SpecimenNumber + SchemaName + by AuthorLastName” (ex: *Davidia antiqua* MLA by Currano). Other acceptable and informative names include: “UCMP 24678 preliminary cuticle description” or “DE 124 morphotype aggregated MLA”. Note that this informative name for a description will be useful when specimens are later re-described or more completely described by another worker. In such a case, the original description (whether simply a text description or scored schema characters) can be retained and a new description node created for the revised descriptions. Both descriptions can be viewed and the historical information is retained alongside newer interpretations.

Schema – Choose the Schema that will be used to make the Description. Only one schema can be used per Description node. If you need to describe a Specimen, or Specimens, using multiple schemas (for example a schema for leaf morphology and a schema for cuticle characters) then you will need to make two Description nodes for the Specimen(s). A Specimen may have multiple Description nodes.

References – Choose an existing Reference, or References, for the Description. If the description record is based on a published description, choose that reference. If not published,

you may choose a personal ‘working reference’ (see Reference Instruction sheet). Enter the order number for the Reference to indicate its priority in the reference list, with the most relevant publication as order 1.

Specimens – Select the Specimen or Specimens that are being described. Add the first specimen by selecting from the drop-down list or by using the search tool. Additional specimens can then be added by clicking “**ADD SPECIMEN**”.

Public or Group – For public data, leave the box checked. For private data, uncheck the box, then from the Groups dropdown select the private data access groups to which this data record belongs. Note that a public Description must have at least one public Specimen. Once a record has been made public, it cannot be returned to a private group.

OPTIONAL FIELDS:

Written description – Paste or type the written description of the Specimen or Specimens. This field allows users to preserve the original terminology used by the describing author, or to provide additional context to the Description that is not possible through the standardized character schema. Because different authors can provide differing descriptions of the same OTU based on specimens from different localities, the written description is recorded for the specific specimens being described, rather than on the OTU itself. Note that the diagnosis text is applied directly to the OTU.

Notes – Enter notes about the Description that the enterer deems necessary to record. For private data undergoing research, this field can be used for research notes or memos to collaborators. Notes can be edited or deleted at any time.

CHARACTER INSTANCES:

The character instances tab will be visible through the “edit Description” function  after a Description node has been submitted. Open the accordion(s) to enter Character States from your chosen Schema. There may be a hierarchical set of accordions depending on how the Schema was designed. If you are scoring character state for a published description, please see p. 53 in the PBot Best Practices section for guidance on translating authors’ text descriptions to standardized schemas.

To record the Character States that comprise your description, click the plus symbol  that is under a Character that you want to record. A pop-up box will appear with a drop-down list of the States available for that Character. Once you select a State, press submit. Each submit creates a “Character instance”.

To record variation in the States exhibited for a particular Character, you may choose to add more States for the same Character. The “Order” field allows you to indicate which of the States is more prevalent by numerically ordering them (note, the order is not an abundance, it is simply a relative ordering). It is possible to give the same order level to multiple states. For example, if state A is dominant and states B, C, and D are equivalent and less common, the user

can apply order level 1 to state A and order level 2 to states B, C, and D. If there is no priority for a particular Character, you may list all States as Order 1. The default is to order the States by the order in which they are entered.

You may edit or delete a character instance using the buttons to the far right   of the character instance.

You do not need to press the gray Submit button at the bottom of the page to save character instances. However, if you edit any Required or Optional fields, do make sure to press Submit.

Operational Taxonomic Unit (OTU) Node

Instructions for data entry

PBot uses the concept of an Operational Taxonomic Unit (OTU) for data entry of fossil taxa. The use of OTUs is a practical one, as it provides flexibility for users by allowing the database to accommodate both formally named Linnaean-style fossil taxa (i.e., Latin binomials), informal taxa such as morphotypes, and even working descriptions (see Cleal and Thomas 2010 & 2021 for nice background on “fossil-taxa” and the International Code of Nomenclature). **In practice here, an OTU should represent a distinct entity that can be defined and differentiated from other fossil taxa based on a set of descriptive features.** As such, an OTU should be considered as the finest-scale taxonomic rank that is possible given the parts preserved and mode of preservation. Ideally, this is at the level of a fossil-species, however with the understanding that taxonomic resolution can differ among plant fossils so that taxonomic rank for different plant parts and preservation modes are not necessarily equivalent. An OTU should not, therefore, be erected in PBot for a taxonomic rank such as family or order when there are described/named taxa at lower rank, although such features may be developed in the future.

REQUIRED FIELDS:

Name – Provide a unique and informative name for the Operational Taxonomic Unit.

- For OTUs that are informally named and include a systematic affinity to a formal name (e.g., Lauraceae sp., *Aralia* sp., cf. *Rhus*), the OTU name should be systematic affinity + morphotype code (if applicable) + locality/flora (e.g., Lauraceae sp. WW036 Bighorn Basin, cf. *Rhus* Bighorn Basin).
- For OTUs that are informally named with no systematic affinity (e.g., Morphotype 27), the OTU name should be the locality/flora + morphotype code or name (e.g., Cerrejón Morphotype CJ38, Mush Long & Loopy, Bighorn Basin Dicot sp. FU735 Hairy Leaf).
- For formally named, published OTUs, the name should simply be the full published name (i.e., the binomial consisting of genus + specific epithet). Note that if you name an OTU with only the formal genus + species epithet in your private workbench, this prevents anyone else from entering the formally named OTU!

Importantly, only use the Linnaean binomial name if you are entering the official taxonomic record (whether original or revised), complete with the published reference, diagnosis, and at least one type specimen. If you are creating an OTU record because you are entering material that is identified as a known taxon that is not yet in the database, then we encourage you to find the original description and type/exemplar specimen and create the public official OTU record for that species. Then you may simply assign the additional new specimens to that taxon. If you are not able to enter the original record for a published/named OTU, as an interim solution you can add a descriptor for your locality to the end of the OTU name (e.g., *Platanites raynoldsii* Big Bluff) to differentiate it from the official OTU record. You may edit/add to this record if you are later able to find and enter the actual diagnosis, reference, locality and type specimen for the official OTU, at which point you can edit the OTU name to essentially convert it to the official record for the taxon. Alternately, if you discover that someone else has entered

the official OTU, then you can reassign your specimens to that OTU node; any Descriptions you have made will carry over, as they are tied to Specimens, not OTUs. Remember to delete your old OTU after reassigning the specimens.

Authority – The authority for the OTU, following ICN rules if the OTU is a validly named taxon or if the OTU is an informal taxon then the authority is the person, or groups of persons, describing the taxon.

Parts preserved – Select the organs or parts of the plant that are described by the OTU diagnosis and description.

Diagnosis – The diagnosis for the taxon. The diagnosis should be a concise statement of the diagnostic features of the taxon that differentiate it from other similar taxa. Note that this field can be used to capture important diagnostic features that may not yet be adequately covered in available character schemas. If the OTU is a published named taxon, paste the verbatim diagnosis from the publication.

Quality index – Choose from High, Medium, Low to indicate the author’s confidence level that this OTU represents a valid distinct taxon. For example, based on the specimens and their preservation, how confident are you that this is a distinct OTU? Are there enough diagnostic characters? Importantly, this field is not meant to convey uncertainty in the taxonomic assignment (e.g., what family/genus/etc.); taxonomic uncertainty should be written into the Notes field with explanation.

We recommend using the Morphotype Quality Index (MQI) of Johnson (1989) and Arens and Allen (2014) to determine the PBot Quality Index as follows:

- *High*: at least 2 well-preserved and/or whole specimens (MQI of 1 or 2)
- *Medium*: one whole or partial well-preserved specimen (MQI of 3)
- *Low*: specimens are poorly preserved; may be one or more whole or partial specimens (MQI of 4 or 5)

The Exemplar Quality Index (EQI) of Wilson et al. 2021 (modified from the holomorphotype quality index of Ash 1999 and Ellis et al. 2009) may be useful to determine what constitutes a “well-preserved” specimen. This metric was developed for ‘dicot’ leaves, but could be easily expanded and applied to other fossil types. Wilson et al. (2021) chose five features to assess the completeness of each specimen (base, apex, margin, third-order venation, fourth-order venation), and the EQI is simply the number of these features present on a specimen (1-5). Wilson et al. (2021) considered a score above 3 to be “well-preserved.”

It is critical for the quality index to be consistent across fossil sites and formations. The highest quality OTU at a site with moderate preservation may only warrant a Medium score for quality index.

Major taxon group – Choose the group to which this OTU belongs from the dropdown. Although many of these are informal groups, they are useful for searching and sorting data within PBot.

PBDB parent taxon – Type the most resolved taxon to which this OTU belongs. This field is used to coordinate taxon records between PBDB and PBot. When you click elsewhere on the page after entering a taxon name, that name will be checked against the taxonomic hierarchy in PBDB; if it does not exist in PBDB you will be notified, at which point you can enter the next higher taxon (or, if you are a registered PBDB data enterer, go add the taxon to PBDB's taxonomic hierarchy). For example, for a named binomial taxon, if the genus or species names are not in PBDB, then you should type the Family name, if known, or next higher order. For informal taxa, use the most resolved higher order taxon possible. In the absolute worst-case scenario for an *incertae sedis* morphotype, this entry would be Plantae. The taxonomic levels above your entered parent taxon will be automatically applied according to PBDB's hierarchy. You can search the PBDB taxonomic hierarchy here: <https://paleobiodb.org/classic/checkTaxonInfo>

References – Choose an existing Reference, or References, for the OTU. For a published taxon, this should be the publication(s) that describe and, if applicable, revise a taxon. For an unpublished taxon, use a personal 'working reference' (see Reference instruction sheet). Enter the order number for the Reference to indicate its priority in the reference list, with the most recent descriptive publication on which the taxon is based as order 1.

Identified specimens – Add one or more Specimens that have been identified as belonging to the OTU. This can include the specimens that were used to originally describe the OTU, as well as specimens later identified as the OTU. Add the first Specimen by selecting from the dropdown list or by using the search tool. Additional Specimens are added by clicking “**ADD IDENTIFIED SPECIMEN**”.

Type specimens – Choose the subset of specimens from the 'Identified specimens' that are considered the type specimens of any sort (holotype, paratypes, syntypes, etc.). For unpublished taxa, these are the exemplar specimens equivalent to type specimens. Note that you can designate which of these is the holotype in the Optional Fields below.

Public or Group – For public data, check the box. For private data, uncheck the box, then from the Groups dropdown select the private data access groups to which this data record belongs. Note that a public OTU must have a public holotype/exemplar specimen. Once a record has been made public, it cannot be returned to a private group.

OPTIONAL FIELDS:

Holotype specimen – Choose the Holotype or “Holo-Exemplar” specimen from the list of type specimens.

Family – Type the family to which the OTU belongs, if known. This is particularly important if the family was not available in the PBDB list and will help with updating the PBDB later.

Genus – Type the genus, if known.

Plant Fossil Names genus link – If available, add the direct link to the genus record in the Plant Fossil Names Registry (<https://www.plantfossilnames.org>) or the International Fossil Plant Names Index (<https://ifpni.org/index.htm>).

Specific epithet – Type the specific epithet, if known.

Plant Fossil Names species link – If available, add the direct link to the species record in the Plant Fossil Names Registry (<https://www.plantfossilnames.org>) or the International Fossil Plant Names Index (<https://ifpni.org/index.htm>).

Additional clades – Type any additional clades or groups (formal or informal) that you would like to record for the OTU. This field allows the database to capture taxonomic information that is not in PBDB. Separate names with a comma.

Notable features preserved – Choose from the list any notable aspects of the OTU that are known to be preserved.

Notes – Enter notes about the OTU that the describer or enterer deems necessary to record. This can include uncertainties about the taxonomy or description, notes on the validity of the taxon or needed revisions that have not been published, etc. In your private access groups, this field can be used for research notes and memos to your collaborators; you may edit or delete such notes before making the OTU record public.

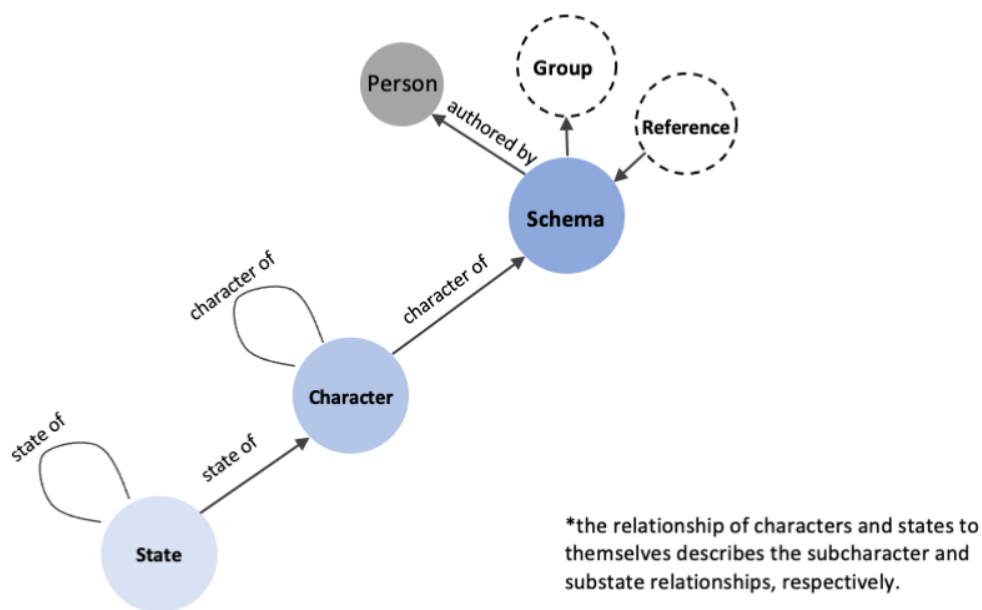
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Schema Node

Instructions for data entry

The primary function of PBot Schemas is to support the development and use of community-sourced descriptions of plant parts (e.g., leaves) and notable features (e.g., cuticles). Each Schema in PBot contains a curated collection of Characters and States, whether qualitative or quantitative, that are intended for a specific purpose (for example, to describe a leaf, or to record insect damage on specimens). We can think of the Characters as questions (e.g., What is the shape of a leaf?) and their States as the answers (e.g., ovate, elliptic, or obovate). Note that States may have substates to record finer scale differentiation of a State when needed or applicable (e.g., in the Manual of Leaf Architecture, the character “epimedial tertiary fabric” can have the state “percurrent” which has the substates “opposite, alternate, or mixed”). Once a Schema has been created, it can be used to describe Specimens, and hence the Taxon (OTU) that a specimen(s) belongs to, so that Specimens and Taxa (OTUs) can be compared. The below diagram illustrates the node-relationship architecture of schemas.



The information provided here is limited; please refer to additional guidance documents for creating and organizing schemas.

REQUIRED FIELDS:

Title – For published schemas, enter the published title. For unpublished schemas, enter a unique and descriptive title related to the purpose of the schema.

Year – For published schemas, enter the copyright date. For unpublished schemas, enter the year the schema was first entered in PBot.

Purpose – What problem is this schema solving? Enter a brief description. Is this a new and unique Schema, an update to an existing Schema, or the first time a published Schema is being

entered into PBot? Is this Schema specific to a particular taxonomic group or time period? Also provide any specific circumstances when it is inappropriate to use this Schema and if an alternate Schema can be suggested, include a recommendation.

Parts preserved – Select parts preserved from the dropdown menu. Multiple parts can be selected at the same time. When selected, parts are highlighted in gray. Click outside of the dropdown menu to return to data entry.

Reference – Whenever possible, select a published Reference. For unpublished schemas, an “unpublished reference” can be used as a temporary placeholder.

Public or Group – Schemas are set to Public, as published Schemas should always be part of the “Public” access group. To restrict access to unpublished Schemas, uncheck “Public.” Then select Group(s) as appropriate from the dropdown menu. Multiple Groups can be selected at the same time. When selected, Groups are highlighted in gray. Click outside of the dropdown menu to return to data entry. Note, once a record has been made public, it cannot be returned to a private group.

OPTIONAL FIELDS:

Acknowledgements – Add acknowledgements that apply to the schema as a whole, just as you would in the acknowledgement section of a publication. Specific citations to acknowledge previously published characters or states can be included in character and state definitions.

Notable features preserved – Select all notable features from the dropdown menu. Multiple features can be selected at the same time. When selected, features are highlighted in gray. Click outside of the dropdown menu to return to data entry.

Character Node

Instructions for data entry

PBot Schemas are hierarchical structures with lower-ranks being dependent on higher-ranks. Therefore, each rank must be added to a Schema in order from the highest position in the hierarchy to the lowest position in the hierarchy.

- PBot instills a hierarchical organization to schemas by allowing users to create Character parent-child relationship between characters; this allows for parent Characters to act as categorical headers as a means to group sets of related characters. The lowest rank Character is the one that should have alternative States.
- PBot allows for up to 5 levels of hierarchical characters (thus up to 4 Heading Characters and then the actual Characters with States). Thus, the Character ranks available in PBot include organizational header, organizational subheader, organizational subsubheader, organizational subsusbsubheader, and character. Although Character is listed as the lowest ranked, it can be dependent on any rank of organizational header. A parent character simply refers to the next higher rank in the hierarchy.

REQUIRED FIELDS:

Schema – Select a schema from the dropdown menu to add a Character.

Parent Character – If applicable, select the higher-ranked organizational header Character that the new Character belongs to. Skip this step if you are entering a top-rank organizational header (and thus not a sub-header). Also skip this step for cases where a character is not dependent on any rank of organizational header. Note that as you add more Characters to a schema, the drop-down menu for Parent Character will reflect the hierarchical organization that has been generated by use of bold font for top-level Characters and then indented sub-headings and Characters.

Name – Enter organizational header names using title case. Enter character names using sentence case. These names should be precise and informative, but concise, as they will appear in the selection drop-down menus when applying a schema to score specimens.

OPTIONAL FIELDS:

Definition – Enter a brief definition using sentence case. Include citation(s) if appropriate.

Order – This function allows schema developers to control where character names appear in lists for users. Enter a number to indicate the position of each character for listing purposes. If no order is specified, PBot will default to alphabetical order. Order is specific to each rank of organizational header and each character rank group (see example below).

Header (order 1): Mammal Coloration
 Subheader (order 1): Mammal Head
 Character (order 1): Nose color

- Character (order 2): Ear color
- Character (order 3): Eye color
- Subheader (order 2): Mammal Tail
 - Character (order 1): Base of tail
 - Character (order 2): Shaft of tail
 - Character (order 3): Tip of tail
- Subheader (order 3): Mammal Body
 - Character (order 1): Primary color
 - Character (order 2): Secondary color
 - Character (order 3): Pattern observed

State Node

Instructions for data entry

PBot schemas are hierarchical structures with lower-ranks being dependent on higher-ranks. Therefore, each rank must be added to a schema in order from the highest position in the hierarchy to the lowest position in the hierarchy. As with Characters, PBot allows a hierarchical parent-child relationship for States. The difference, however, is that each lower rank of a state is an increasingly precise answer to the question – it is not simply an organizational tool.

- There are up to five state ranks available in PBot, thus allowing for a state, substate, subsubstate, subsubsubstate, and subsubsubsubstate.
- A lower-ranked state such as subsubstate is directly dependent on a higher-ranked state such as substate. When dependencies exist, the higher-ranked state is known as the parent state.

REQUIRED FIELDS:

Schema – Select a Schema from the dropdown menu to add an associated State.

Character – Select the Character from the dropdown menu to which the State being added belongs.

Parent state – If the new state is dependent on another state (i.e., it is a substate), then select the higher-ranked state that the new state being added is dependent on. Skip this step if you are entering a state that is not dependent on another state.

Quantitative – Check this box if the state being entered is quantitative. If the state is categorical, leave this box unchecked.

Name – Enter state name using lower case (e.g., marginal). Do not include terms like “unknown” or “indeterminate” as states. These can lead to erroneous similarities in search or comparison algorithms.

OPTIONAL FIELDS:

Definition – Enter a brief definition using sentence case. Include citation(s) if appropriate.

Order – This function allows schema developers to control where “state” names appear in lists for users. Enter a number to indicate the position of each “state” for listing purposes. Order is specific to each character and each “state” rank group (see example below).

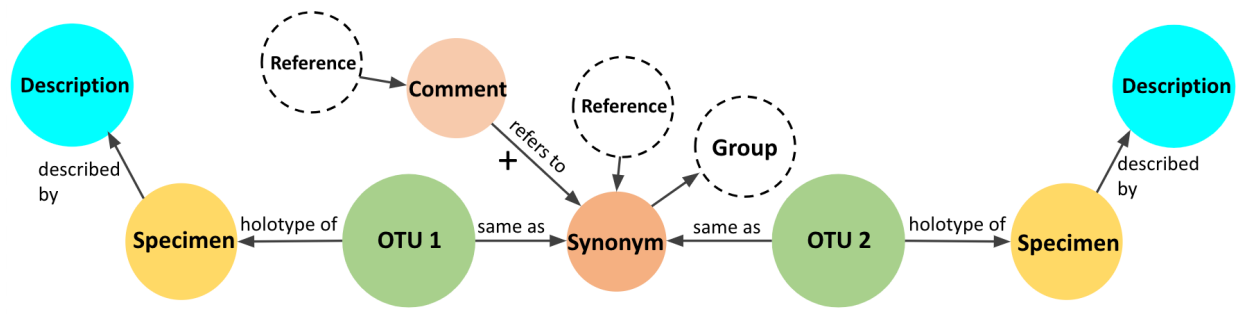
Character (order 1): Primary color of the mammal body
state (order 1): white
substate (order 1): solid
substate (order 2): spotted
substate (order 3): striped
substate (order 4): other

state (order 2): black
 substate (order 1): solid
 substate (order 2): spotted
 substate (order 3): striped
 substate (order 4): other
state (order 3): tawny
 substate (order 1): solid
 substate (order 2): spotted
 substate (order 3): striped
 substate (order 4): other
state (order 4): dark brown
 substate (order 1): solid
 substate (order 2): spotted
 substate (order 3): striped
 substate (order 4): other

Synonym Node

Instructions for data entry

One of our favorite PBot capabilities is being able to compare OTUs from different sites, time periods, and research groups. And in doing so, it is possible to recognize when different names are being used to refer to the same taxon and to propose **synonymies** between OTUs. We use the term **Synonymy** in a slightly less formal sense than the strict taxonomic definition, but the essence is the same- to denote a relationship between OTUs that were given different names but are thought to represent the same taxon. The Synonymy feature is a work in progress and will undergo significant upgrades in our next funding cycle.



The current implementation of the “synonymy” feature creates a link between two or more existing OTUs and allows the enterer to add comments explaining why the OTUs are being linked and add supporting references for those actions. The OTUs that are linked retain their independent OTU pages, and all specimens remain assigned to their original OTU. The proposed synonymy displays on each of the OTU's pages. Any user can add additional tracked comments to the proposed synonymy, or existing comments, to facilitate dialog.

What it DOES NOT do: Proposing a synonymy does not merge the data from the linked OTUs. No data is lost or irreversibly migrated by a synonymy event.

WHEN TO SYNONYMIZE:

In the current PBot functionality, information about Synonyms can be viewed at the bottom of the OTU page, under the Synonym accordion. However, the synonym relationship is not incorporated into search functions yet (this is one of our top priorities for PBot v2.0). For example, if you search for all OTUs from a time interval, you will see both OTUs listed separately. Given the current constraints of PBot, we provide these recommendations for when to create distinct OTUs and a synonym relationship between them, versus assigning specimens to an existing OTU:

- If you are very confident that a specimen you are entering belongs to an OTU that already exists in PBot, we strongly encourage you to assign that specimen to the existing OTU, rather than create a new OTU and a Synonym. To do this in your workbench, navigate to OTU entry form, select the edit icon, and then search and select the existing OTU. Add your new specimen(s) as an “Identified specimen” and press submit. Now

your new specimens will appear on the OTU page as additional occurrences of the taxon.

- If you are morphotyping a flora and have created two or more distinct OTUs that you later decide to lump together prior to publishing your work (and making your OTUs “public” in PBot), we strongly encourage you to reassign specimens such that only a single OTU exists. To do this, edit the OTU page for the OTU name that you plan to retain, adding all specimens as an “Identified specimen.” Then, delete the other OTU nodes.
- For published taxa that have been formally synonymized in the literature, particularly for long-accepted synonymizations, we suggest editing the affected OTUs to reflect the updated literature and to reassign affected specimens to the OTU that will be conserved in PBot (this would general be the one with taxonomic priority). To do this in your workbench, choose to edit the OTU that will be conserved, and edit its name, diagnosis, and text descriptions as needed, and check that the type specimens and other metadata and taxonomic fields are correct. Add the reference for the synonymy as the primary reference and explain the taxonomic history in the notes field. For the OTU(s) that are being synonymized, navigate to their pages to check that all pertinent data (such as in the notes field) has been copied to the priority OTU, and that all specimens have been reassigned, and then you may delete the now-defunct OTU.
- In the case that you or another author is proposing that two published OTUs are the same, but the synonymization has not yet been validly published, simply create a Synonym node and provide the explanation for the proposed synonymy. Do not re-assign specimens or delete OTUs (unless the affected OTUs are part of your own unpublished work).
- For whole-plant concepts that consist of multiple organ-specific named taxa, we recommend creating/maintaining separate OTUs for each formal taxonomic name (or organ) and synonymizing them. This allows other single-organ occurrences to still be assigned to the appropriate organ-specific OTU.
- Currently in PBot, we recommend that OTUs named or identified using open nomenclature, such as “cf.”, “aff.”, or “sp.”, simply be entered as a unique OUT. It is up to the user’s discretion as to whether it is appropriate to create a proposed synonymy between such OTUs and established taxa.

REQUIRED FIELDS:

Explanation – A concise text explanation for the proposed synonymy.

OTUs – Select two OTUs that are proposed to be synonymous. Both OTUs must already exist in PBot. Currently, it is only possible to link two OTUs at a time, and so if a Synonym is proposed that links more than two OTUs, it is necessary to create pairwise links between all OTUs.

Reference – Select one or more References that state that the two OTUs represent a single entity and should be synonymized. References must already exist in PBot and can include unpublished references (temporary placeholders for personal workbenches and unpublished opinions). Multiple References can be associated with a Collection by clicking “**ADD**”

REFERENCE". Enter the order number for the Reference to indicate its priority in the reference list, with the most relevant publication as order 1.

Public or Group – The default is for a Synonym to be Public, as published data and taxonomic opinions should always be part of the "Public" access group. If you would like to restrict access to an unpublished Synonym, uncheck "Public" and select which one or more of your private groups can access this Synonym. Once a record has been made public, it cannot be returned to a private group.

OPTIONAL FIELDS—not applicable

Comment Node

Instructions for data entry

Currently, the Comment node can only be used for Synonyms, although we hope to add this feature to other node types in the future. The purpose of this node is to provide feedback on proposed Synonyms.

REQUIRED FIELDS:

Synonym – Choose the Synonym node that you would like to write a Comment about.

Comment – Write out the text of your comment. Include the date at the end of the text. There is no need to include your name, as it will be automatically displayed with the Comment.

Parent Comment – If you are replying to an existing comment, choose that comment from this list. If you would like to start a new comment thread on the Synonym, then leave this field blank.

OPTIONAL FIELDS:

Reference – Select one or more References that support your Comment, if applicable. Only include published References, and these must already exist in PBot. Multiple References can be associated with a Comment by clicking “**ADD REFERENCE**”. Enter the order number for the Reference to indicate its priority in the reference list, with the most relevant publication as order 1.

PBot Best Practices

The following sections provide guidance on how to enter data into PBot following best practices that have been developed to maximize the informativeness of entered data, encourage standardization across datasets, and handle a variety of situations that may be encountered when entering data that has been published by other authors, particularly older works. Additionally, we have provided tips for making data entry faster based on our experiences.

How to name nodes

PBot requires all node names to be unique. Below, we provide regulations and guidelines for naming the different node types.

- **Group Name:** Group names should be short and specific to the purpose of the group (e.g., Bighorn Basin P-E).
- **Collection Name:** Choose a unique identification that may aid in recognizing close links among Collections. For example, if you are entering a series of consecutive beds, you could provide a location name and meter level (e.g., “Saddle Mountain 1.5 meter bed” and “Saddle Mountain 2.0 meter bed”). Names must be under 80 characters for compliance with the PBDB.
- **Specimen Number:** When entering the “Specimen number,” write the repository code before the repository’s catalog number if it is not already part of the catalog number. For example, a specimen at the Smithsonian National Museum of Natural History should begin with USNM followed by the catalog number, resulting in a unique Specimen number “USNM PAL777113”. Do not forget to include the repository code, as different institutions may use the same numbering system and Specimen numbers must be unique in PBot. If a catalog number has not been assigned, a field number or other tracking may be temporarily used and edited at a later time.
- **Description Name:** For published OTUs/Specimens, we recommend the format “OTUName/SpecimenNumber + Author(s) + SchemaName + Published Description” (ex: *Davidia antiqua* Manchester MLA Published Description). For unpublished work or new descriptions of published OTUs/Specimens, we recommend the format “OTUName/SpecimenNumber + SchemaName + by AuthorLastName” (ex: *Davidia antiqua* MLA by Currano). Other acceptable and informative names include: UCMP 24678 preliminary cuticle description or DE 124 morphotype aggregated MLA. Note that this informative name for a description will be useful when specimens are later re-described or more completely described by another worker. In such a case, the original description (whether simply a text description or scored schema characters) can be retained and a new description node created for the revised descriptions. Both descriptions can be viewed and the historical information is retained alongside newer interpretations.
- **OTU Name:** OTU names should be unique and informative.
 - For OTUs that are informally named and include a systematic affinity to a formal name (e.g., Lauraceae sp., *Aralia* sp., cf. *Rhus*), the OTU name should be systematic affinity + morphotype code (if applicable) + locality/flora (e.g., Lauraceae sp. WW036 Bighorn Basin, cf. *Rhus* Bighorn Basin).

- For OTUs that are informally named with no systematic affinity (e.g., Morphotype 27), the OTU name should be the locality/flora + morphotype code or name (e.g., Cerrejón Morphotype CJ38, Mush Long & Loopy, Bighorn Basin Dicot sp. FU735 Hairy Leaf).
- For formally named, published OTUs, the name should simply be the full published name (i.e., the binomial consisting of genus + specific epithet).

Importantly, only use the Linnaean binomial name if you are entering the official taxonomic record (whether original or revised), complete with the published reference, diagnosis, and at least one type specimen. If you are creating an OTU record because you are entering material that is identified as a known taxon that is not yet in the database, then we encourage you to find the original description and type/exemplar specimen and create the public official OTU record for that species. Then you may simply assign the additional new specimens to that taxon. If you are not able to enter the original record for a published/named OTU, as an interim solution you can add a descriptor for your locality to the end of the OTU name (e.g., *Platanites raynoldsii* Big Bluff) to differentiate it from the official OTU record. You may edit/add to this record if you are later able to find and enter the actual diagnosis, reference, locality and type specimen for the official OTU, at which point you can edit the OTU name to essentially convert it to the official record for the taxon. Alternately, if you discover that someone else has entered the official OTU, then you can reassign your specimens to that OTU node; any Descriptions you have made will carry over, as they are tied to Specimens, not OTUs. Remember to delete your old OTU after reassigning the specimens. Note that if you name an OTU with only the formal genus + species epithet in your private workbench, this prevents anyone else from entering the formally named OTU!

How to enter published data

You who are about to enter published data, we salute you! Thank you for making these data open and accessible to the community! (Remember, published data should always be assigned to the Public group.) Recently published papers should be straightforward and relatively easy to enter, but entering data that predates GPS units, morphotype systems, and descriptive schemas, to give three examples, can pose challenges. Below, we provide best practices for entering published data, based upon our own experiences. The information given below is ordered based on the order of PBot entry.

At least one specimen is required to create an OTU. How do I choose the best exemplar (preferably a type) specimen(s) to enter?

To some extent, the answer to this question depends on what information is available for each specimen listed as type material and/or figured in a publication. However, we recognize that the information needed to create a Specimen node (and the accompanying Collection node) may be difficult to impossible to find in papers published a long time ago.

- Ideally, you should enter the primary exemplar specimen used in the published description. The order of preference is holotype or lectotype or neotype > syntype > paratype > hypotype. You should choose a figured specimen.

- If there are other figured specimens that you would like to feature in the images of the OTU, you must enter those specimens in addition to the primary exemplar. This will allow you to add those images later.
- If the OTU has already been entered in the [Plant Fossil Names Registry](#) (PFNR) or the [International Fossil Plant Names Index](#) (IFPNI), then much of the information you need may be easily found there. If not, consider entering formally named taxa there, as well as in PBot!
- Remember, you need to find and enter the locality data (i.e. Collection node) for the specimen before you can actually enter the specimen.

What if there is poor locality information? (i.e., it's difficult to create the Collection node)

Do your best to georeference the site, recognizing that this can be really difficult for fossils collected a long time ago, and use the “GPS coordinate uncertainty” field to account for the uncertainty in the location. For example, if all you know is that the specimen is from the U.S. state of Wyoming, use Google Maps or another resource to determine the latitude and longitude of the approximate midpoint of Wyoming. Then set the GPS coordinate uncertainty to the approximate radius of Wyoming (300,000 meters).

Sometimes you can find the record(s) for the type specimen(s) in the repository's publicly accessible collections database, and we have found cases where the locality information, including gps coordinates and uncertainty, is more complete in the museum's records than in the original publication. It is a good practice to look for those online records to cross reference information for the Collection node, and for adding a link to the repository's record in the Specimen node. Make sure to indicate in the notes fields what information came from specimen metadata rather than directly from the original publication.

If the paper does not include any lithological data about the site, choose “Sedimentary” for the lithology field.

What if the taxonomy is outdated?

Prior to the use of morphotypes, paleobotanists commonly assigned fossil specimens to modern genera, without the data to back it up. This is particularly problematic for otherwise outstanding monographs published in the late 1800's and early 1900's. (For the North American dicot workers, we provide the examples of Knowlton, Lesquereux, Berry, Brown, etc). Here, we list best practices for creating OTU nodes from historic publications.

- If the generic assignment is questionable (e.g., the many OTUs that were assigned to the modern genus *Ficus* in the early paleobotanical literature), place the genus name in quotes (ex: “*Ficus*” obovata). Ideally, there should be a published reference supporting the use of quotation marks and this reference should be entered into PBot and listed as a reference supporting the OTU. If the generic assignment has not yet been directly questioned in the literature, but you feel strongly that this questionable assignment should be noted, we suggest the following: (1) provide a written explanation in the notes field, (2) use quotation marks around the genus only in cases where a living genus was used and likely inappropriate. PBot is not intended to override what is validly published,

yet we also want to prevent confusion by workers that may be more unfamiliar with the history of paleobotanical naming practices and evolving systematics that are trying to use fossil data to inform other analyses (for example, in phylogenetic analyses).

- What do I enter for the taxonomic information? For most cases, in the PBDB parent taxon field, you will choose the higher rank parent taxon that is explicitly stated in the published taxonomic treatment (even if you believe it is incorrect and needs revision), and then enter the published taxonomic information (family, genus, specific epithet) into the appropriate fields in the Optional fields accordion menu. You may use quotes in the free text fields to indicate clearly questionable assignments as mentioned above. In the Notes field, please include any taxonomic opinions from the published description or subsequent works. You may also record your comments on the taxonomy here to aid future users or workers. Hopefully, recording such opinions and dialog will help prompt the needed formal revisions. We strongly suggest to check the [Plant Fossil Names Registry](#) (PFNR) and the [International Fossil Plant Names Index](#) (IFPNI) for the nomenclatural status of the scientific names you are entering. These two resources are actively building authoritative registries of fossil plant names based on published literature and contain helpful information on nomenclatural acts, typification, and references, etc.

How do I enter a published description?

There are three key aspects of the description node architecture that are important to understand in order to make best use of the PBot's capabilities for handling different scenarios.

1. Descriptions are applied to one or more specimens, not directly to a taxon. This is because specimens are the primary data that the description is based upon, and associating descriptions with the specimens themselves allows for specimen-level variation in traits to be recorded and employed in useful ways. It is also practical; for example, if a specimen is re-assigned to a different OTU, the specimen's description transfers with it.
2. A specimen, or group of specimens, can have multiple description nodes. This allows users the flexibility to add multiple interpretations of specimens by different authors, as well as apply multiple different types of descriptions using different schemas to specimens (as explained further below).
3. Description nodes are schema-specific; therefore, you may create more than one description node to describe features of the fossil specimens that are circumscribed by separate schemas (for example, a leaf's morphology from the Manual of Leaf Architecture schema and the functional physiognomic traits from that leaf using the Dicot Quantitative schema).

Note that these three aspects in combination can account for a wide range of case scenarios. For example, a group of specimens could have a shared description of leaf morphology based on the Manual of Leaf Architecture schema. One specimen in this group has cuticle preserved and therefore has an additional description using the Cuticle Schema, while a different subset of specimens have been measured for functional traits and recorded via a the Dicot Quantitative schema.

In practice, you should create separate description nodes for each schema that you plan to use and for each reference or person who described the specimen(s). Make sure the relevant specimens and references are entered before attempting to create a description.

- First, create a description node for a specific schema, named following the conventions in the node naming guidelines above. This description node will contain the relevant part of the description from the publication, with as little interpretation as possible.
 - Choose the relevant schema for this description node (e.g., MLA for dicot leaves), select the published reference, and select the specimens for which this description applies. We encourage you to list all Specimens that you were able to enter from the publication in the entry form for this Description.
 - Copy the publication's description (or the diagnosis if only a diagnosis is given) verbatim and put this in the "Written description" optional field. Only copy the relevant part of the description for the schema chosen to describe the specimen(s) (for example the leaf part of the description for a leaf specimen described by the MLA).
 - After creating the description node, go back and edit that description node to add the character instances for characters that were explicitly stated in the publication, minimizing interpretation as much as possible. To stay faithful to the reference that you are entering, this will sometimes mean that you cannot score every character that you can see and interpret on the specimens (whether via the specimen figures or on the actual specimens).
 - If you would like, you can create an additional description node in which you describe, based on your own observations or interpretations, one or more specimens featured in the publication. This is a great way to update descriptions that might not use modern terminology or conventions while separately maintaining the integrity of the original author's interpretations. Name this node "OTUName SchemaName by YourLastName" (ex: *Davidia antiqua* MLA by Currano). You may choose to assign this description to a private group, particularly if you are working on a taxonomic revision, but remember to make the description public upon publication.
- Repeat this process for parts of the published description that are described by other schemas.

Diagnoses vs. Descriptions

According to the *International Code of Nomenclature for Algae, Fungi, and Plants*, "A diagnosis of a taxon is a statement of that which in the opinion of its author distinguishes the taxon from other taxa." A description is defined as "a published statement of a feature or features of an individual taxon... a validating description need not be diagnostic." Ideally, publications will have separate sections labeled "diagnosis" and "description." If this is the case, then the publication's diagnosis section should be copied verbatim into the diagnosis field in the OTU entry form, and the relevant text from the publication's description section should be copied verbatim into the "written description" optional field in the Description entry form (see below). If the publication only has a description section or only has a diagnosis section, then the text

there should be copied verbatim into both the diagnosis field in the OTU entry form and the “written description” optional field in the Description entry form.

What if there is no appropriate schema in PBot to describe my fossils?

If there is not an appropriate schema available that applies to the organ/preservation that you are describing, then you may do one of the following. (1) select the “General text description only schema” and enter the description in the “Written description” field. No characters or states exist in this schema. (2) Create a custom schema for describing your fossils. Note that your schema will need to be made “public” in order for users to see your description. Anyone may use a public schema, and in this way PBot may be able to facilitate building schemas for description of different aspects of fossil plants.

How can I include images/illustrations of specimens?

First, check to see if the specimen’s repository has an online image of specimens. If so, enter the stable URL for that image. If not, take a screenshot from the publication and upload that to the PBot Internet Archive collection or upload directly to PBot. And if you should happen to visit the specimen, please take a great photograph and create a new Image node. Note that PBot has a limit on the size of images uploaded. If you encounter this problem, you must downsize the image or host the image on a different site instead and link to it from PBot. Make sure you have permission to upload these images or that any images that you are linking or uploading are public domain! Many online collection databases and journals hold copyright over their specimen photographs.

PBot Pro Tips

- Open separate windows for the Workbench and Explorer to be able to immediately view the changes you make. This is especially helpful for entering Schemas.
- If you are midway through a form and realize that you forgot to enter a node that is required for your form (e.g., a Person or Reference), simply open another window and enter that node. It will then populate in the search in the form that you had started.
- As more and more data are entered into PBot, the dropdown menus will take longer and longer to load. The Specimen dropdown menus are already quite slow to load! To avoid slow loading or a long scrolling list, use the magnifying glass icon to search for nodes instead of the dropdown selection on entry fields.
- Whenever searching PBot (via PBot Explore or by clicking the magnifying glass when entering data in a form), make sure that you are selecting all groups that you want to search. The default is to only search public, and you will not get results from your private groups.
- Command-Shift-V or Control-Shift-V allows you to paste copied text without formatting. This can be especially useful for removing line breaks from diagnoses in old pdfs.
- Pushing return while in a data entry field will submit the form.
- Groups and References can be useful for project management purposes. You can search Collections, Specimens, and OTUs by Group or by Reference, which can be very useful for keeping track of progress.

- PBot Explore's default is to only search Public records. Remember to open the Metadata accordion and add private Groups to search both public and private data.